

Mathematics Stage 4

# Properties of Geometrical Figures

Book 1

Version: 261002

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# Syllabus Content

**MA4-GEO-C-01** identifies and applies the properties of triangles and quadrilaterals to solve problems

## **Classify triangles according to their side and angle properties**

- Label triangles using appropriate text and symbols
- Classify and describe types of triangles based on their properties, including acute-angled, right-angled, obtuse-angled, equilateral, isosceles and scalene triangles

## **Classify quadrilaterals and describe their properties**

- Identify quadrilaterals using naming conventions
- Distinguish between convex and non-convex quadrilaterals
- Verify and describe the properties of the special quadrilaterals which include parallelograms, rectangles, rhombuses, squares, trapeziums and kites
- Identify and label the properties of the special quadrilaterals using appropriate conventions
- Classify quadrilaterals based on their properties
- Justify why some quadrilaterals may be classified as more than one type of quadrilateral

## **Apply the properties of triangles and quadrilaterals**

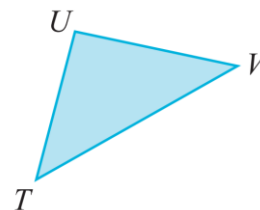
- Prove that the interior angle sum of a triangle is  $180^\circ$  with or without digital tools
- Prove that any exterior angle of a triangle equals the sum of the 2 interior opposite angles
- Apply the angle sum of a triangle to prove that the angle sum of a quadrilateral is  $360^\circ$
- Apply the properties of triangles and quadrilaterals to determine unknown sides and angles to solve problems, giving reasons

# Classifying Triangles

## i Labelling Triangles

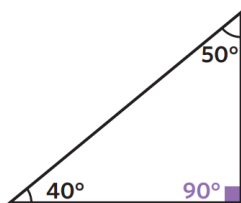
- We label triangles according to its three vertices.
- Name the vertices in clockwise order.

$\Delta TUV$



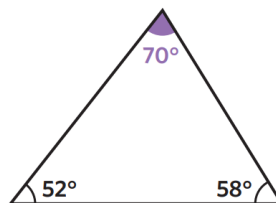
## i Classifying Triangles According to Angle Properties

### Right-Angled Triangle



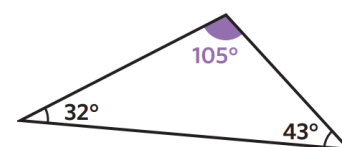
One angle is  $90^\circ$

### Acute-Angled Triangle



All angles acute ( $<90^\circ$ )

### Obtuse-Angled Triangle

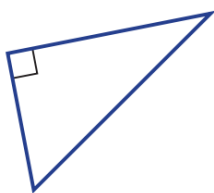


One angle obtuse ( $>90^\circ$ )

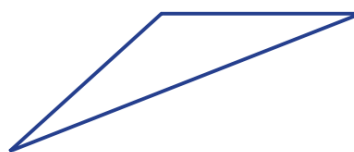
## Q Classify triangle based on angle property

Classify these triangles as right-angled, acute, or obtuse.

a



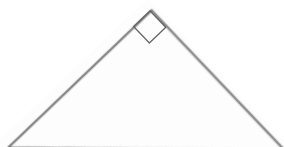
b



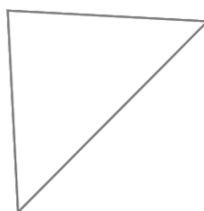
c



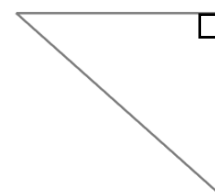
d



e

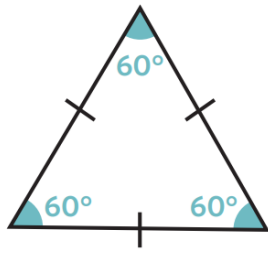


f



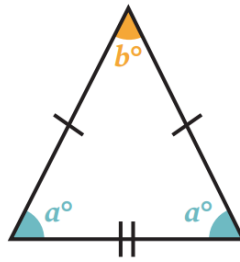
## i Classifying Triangles According to Angle Properties

### Equilateral Triangle



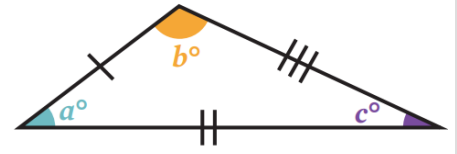
All sides the same length  
All angles the same ( $60^\circ$ )

### Isosceles Triangle



Two side lengths equal  
Two angles the same

### Scalene Triangle

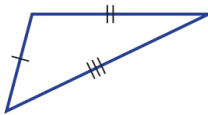


All three sides different.  
All angles different.

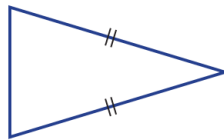
## Q Classify triangle based on side property

Classify these triangles as equilateral, isosceles, or scalene.

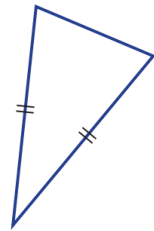
a



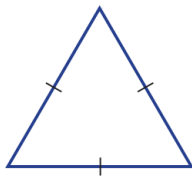
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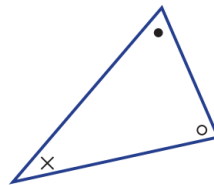
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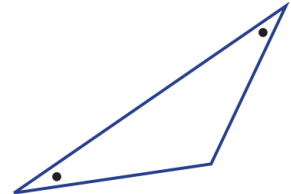
d



e

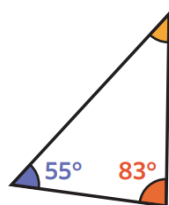


f



1 Classify these triangles. Circle every description that matches.

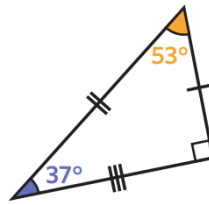
a



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Scalene, acute

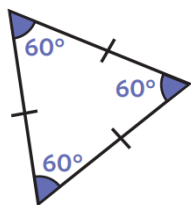
b



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Scalene, right

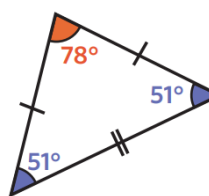
c



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Equilateral, acute

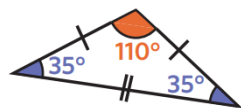
d



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Isosceles, acute

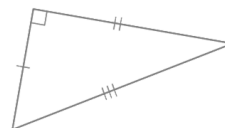
e



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Isosceles, obtuse

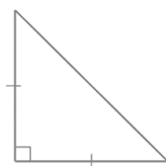
f



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Scalene, right

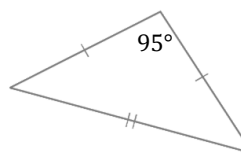
g



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Isosceles, right

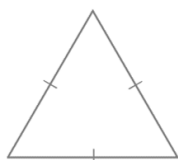
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Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Isosceles, obtuse

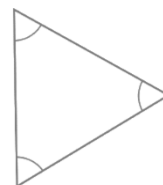
i



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Equilateral, acute

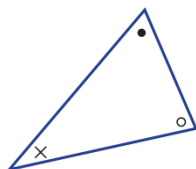
j



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Equilateral, acute

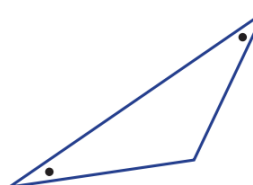
k



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Scalene, acute

l



Equilateral  
Isosceles  
Scalene  
Acute  
Right  
Obtuse

Isosceles, obtuse

2 Explain why every equilateral triangle is also an acute triangle.

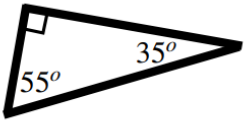
.....  
.....

Equilateral triangles have all angles  $60^\circ$ , so it is an acute triangle.

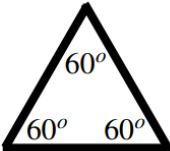
3 Categorise each triangle below into one of the table cells.

|        | Isosceles | Equilateral | Scalene |
|--------|-----------|-------------|---------|
| Acute  | C         | I, B        | E, G    |
| Right  | F         |             | A       |
| Obtuse | D         |             | H       |

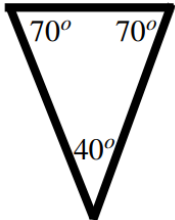
a



b



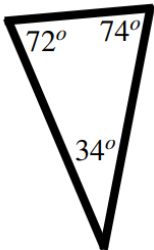
c



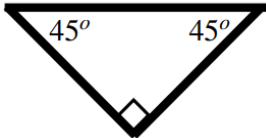
d



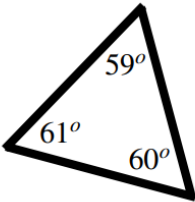
e



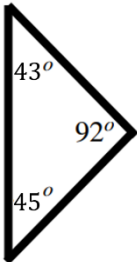
f



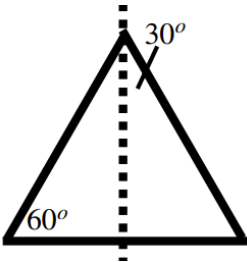
g



h



i



a Explain why the ‘equilateral column’ has two blank boxes.

All equilateral triangles are acute.

b Identify the type of triangle based on the following information:

- i

A triangle with all sides and angles equal

.....

Equilateral
- ii

A triangle with no sides equal

.....

Scalene
- iii

A triangle with one angle 91

.....

Obtuse
- iv

A triangle with angles 103, 20, 57

.....

Scalene, obtuse
- v

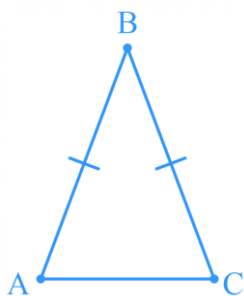
A triangle with sides 11, 15, 11

.....

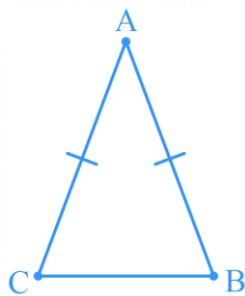
Isosceles

4 Draw the triangles given by these descriptions,

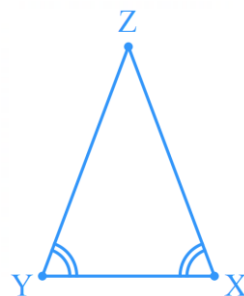
**a**  $\triangle ABC$  isosceles  
with  $AB = BC$



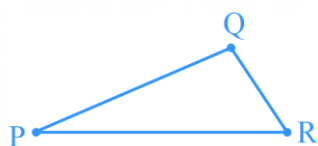
**b**  $\triangle ABC$ , isosceles  
with  $AB = AC$



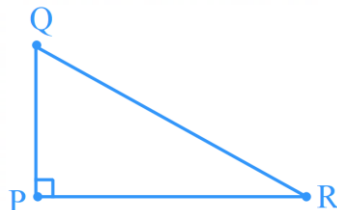
**c**  $\triangle XYZ$ , isosceles,  
 $\angle ZXY = \angle ZYX$



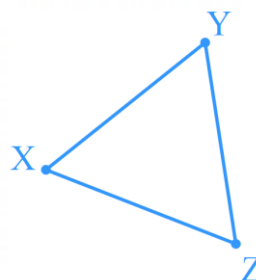
**d**  $\triangle PQR$ , scalene  
not right-angled



**e**  $\triangle PQR$ , scalene with a  
right angle at  $\angle RPQ$



**f**  $\triangle XYZ$ , equilateral

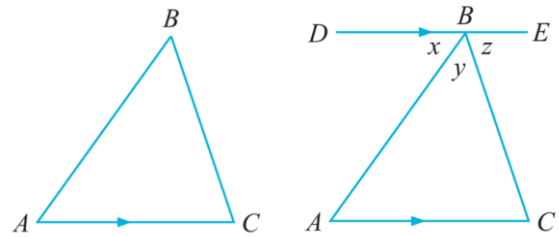


# Angle Sum of a Triangle

## Q Angle Sum of a Triangle

Consider the triangle  $\triangle ABC$ :

Construct line  $DBE$  parallel to  $AC$ .



By considering  $\angle DBE$ :  $x + y + z = \dots\dots\dots^\circ$  ( $\dots\dots\dots$ )

$\angle DBA = \angle BAC$  ( $\dots\dots\dots$ )

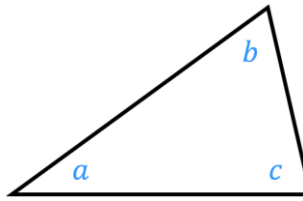
$\angle EBC = \angle \dots\dots\dots$  (alternate angles on parallel lines)

By considering the sum of the angles in  $\triangle ABC$ :

$x + y + z = \dots\dots\dots^\circ$

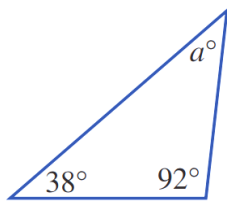
## i Angle Sum of a Triangle

$$a + b + c = 180^\circ$$

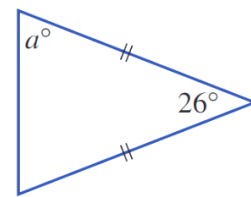


1. Write an equation using angle relationships.
2. Simplify the equation, if needed.
3. Solve for the unknown using inverse operations.

## Example Solve problems involving angle sum of a triangle



1.  $a + 38 + 92 = 180$  (angle sum of triangle)
2.  $a + 130 = 180$
3.  $a = 50^\circ$

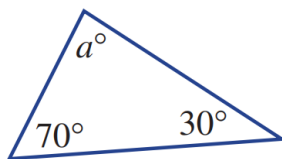


1.  $a + a + 26 = 180$  (angle sum of triangle)
2.  $2a + 26 = 180$
3.  $2a = 154$   
 $a = 77^\circ$

## Guided Practice

Find the value of the pronumerals in these diagrams.

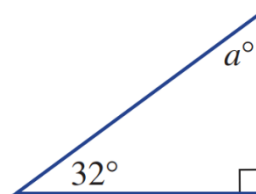
a



$$a + 70 + 30 = 180$$

$$a = 80^\circ$$

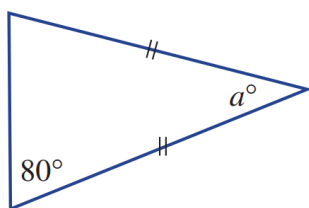
b



$$\dots + \dots + \dots = 180$$

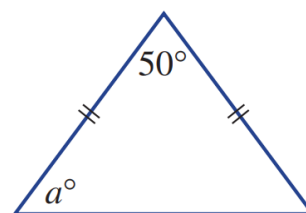
$$a = 58^\circ$$

c



$$a = 20^\circ$$

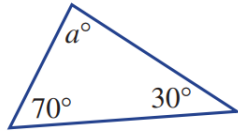
d



$$a = 65^\circ$$

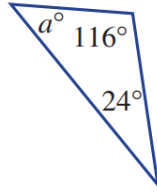
1 Use the angle sum of a triangle to find the value of the pronumeral. Show full working out.

a

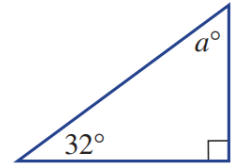


$$a + 30 + 70 = 180$$

b



c

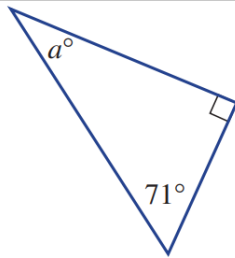


$$a = 80^\circ$$

$$a = 40^\circ$$

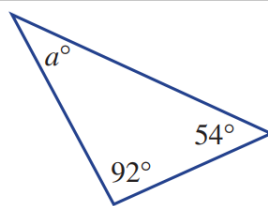
$$a = 58^\circ$$

d



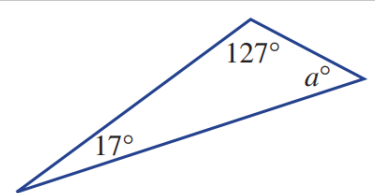
$$a = 19^\circ$$

e



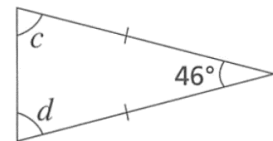
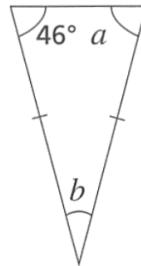
$$a = 34^\circ$$

f



$$a = 36^\circ$$

2 Both of the triangles shown are isosceles.



a What part of the diagram tells you the triangles are isosceles?

The dashes

b Determine the value of the missing angles  $a, b, c, d$ .

$$a = 46^\circ$$

$$b = 88^\circ$$

$$c = d = 67^\circ$$

c Explain why angles  $a$  and  $b$  are different to  $c$  and  $d$ , even though both triangles have similar information.

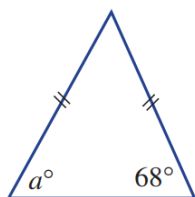
In the first triangle,  $46^\circ$  was one of the equal angles.

In the second triangle,  $46^\circ$  was the non-equal angle.

## DEVELOPMENT

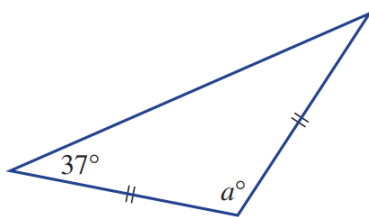
- 3 Use the angle sum of a triangle to find the value of the pronumeral in these isosceles triangles. You do not need to show working out.

a



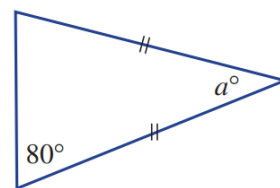
68°

b



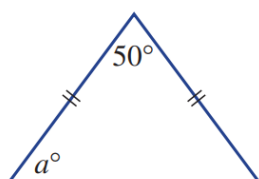
106°

c



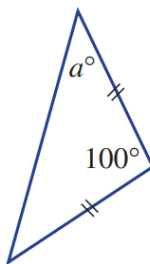
20°

d



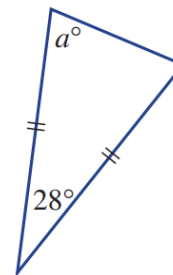
65°

e



40°

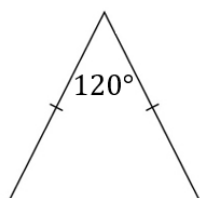
f



76°

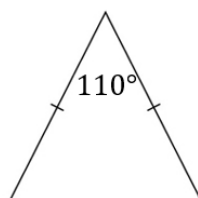
- 4 Determine the value of the unknown angles in these triangles. You do not need to show working out.

a



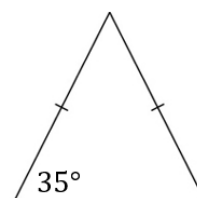
30, 30

b



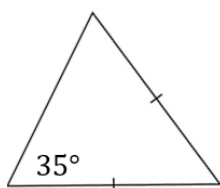
35, 35

c



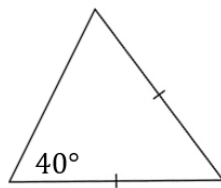
110, 35

d



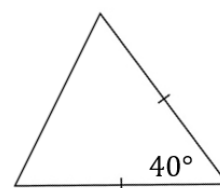
35, 110

e



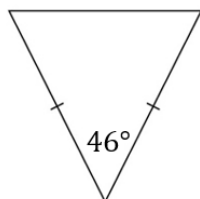
40, 100

f



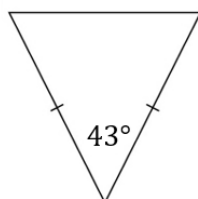
70, 70

g



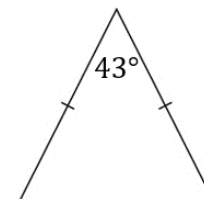
67, 67

h



68.5, 68.5

i



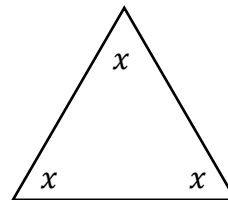
68.5, 68.5

- 5 Two of the angles in a triangle are  $74^\circ$  and  $104^\circ$ . Work out the size of the third angle.

$$a = 74 + 74 = 180^\circ$$

$$a = 2^\circ$$

- 6 Use algebra to explain why every angle in an equilateral triangle is  $60^\circ$ .



$$3x = 180^\circ$$

$$x = 60^\circ$$

- 7 One of the angles in an isosceles triangle is  $80^\circ$ . Work out the sizes of the other two angles.

If  $80^\circ$  is the non-equal angle, then the other angles are  $50^\circ$

If  $80^\circ$  is one of the equal angles, then the other angle is  $20^\circ$

Explain why there are two possible sets of answers.

The question doesn't specify whether  $80^\circ$  is one of the equal angles or not

- 8 One of the angles in an isosceles triangle is  $160^\circ$ . Work out the sizes of the other two angles.

Other two angles are both  $10^\circ$

Explain why there is only one possible set of answers.

$160^\circ$  can't be one of the equal angles because it would exceed the angle sum of a triangle

- 9 An isosceles triangle has an interior angle of  $60^\circ$ . Find all the other angles.

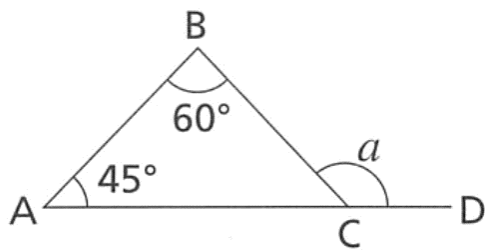
The other two angles are  $60^\circ$

What do you notice?

If an isosceles triangle has  $60^\circ$ , it must be equilateral

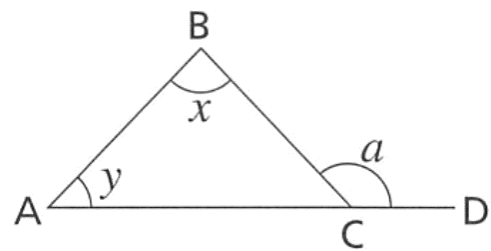
10 Provide full reasoning for both parts below.

a Show that  $a = 105^\circ$



$\angle ACB = 75^\circ$  (angle sum of a triangle)  
 $a = 180 - 75 = 105^\circ$   
(angles on a straight line supplementary)

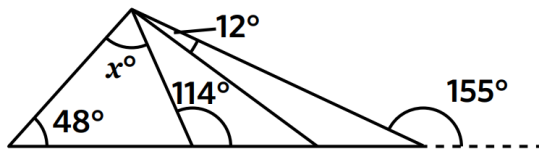
b Prove that  $a = x + y$



$\angle BCA = 180 - x - y$  (angle sum of triangle)  
 $a = 180 - (180 - x - y)$  (angles on a straight line)  
 $a = x + y$

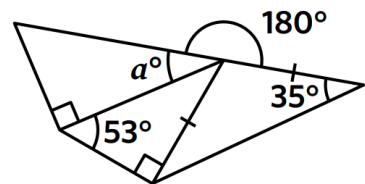
11 Determine the value represented by the pronumeral. You do not need to show working out.

a



$x = 66^\circ$

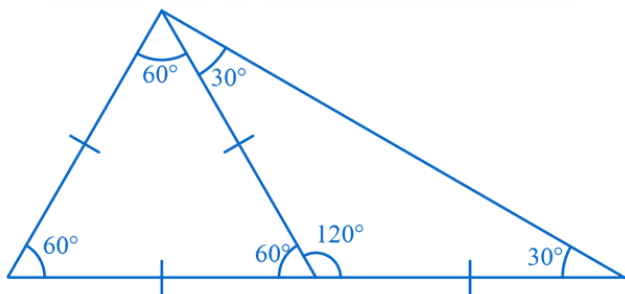
b



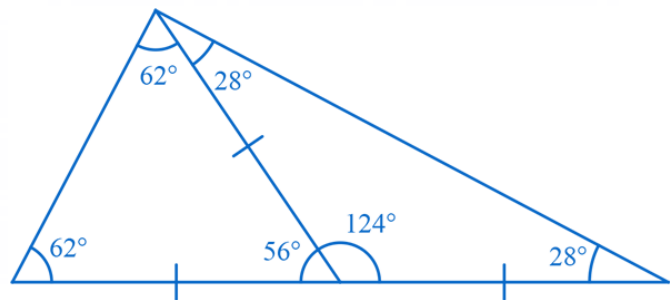
$a = 33^\circ$

12 Determine the size of the missing angles in each diagram. You do not need to show working out.

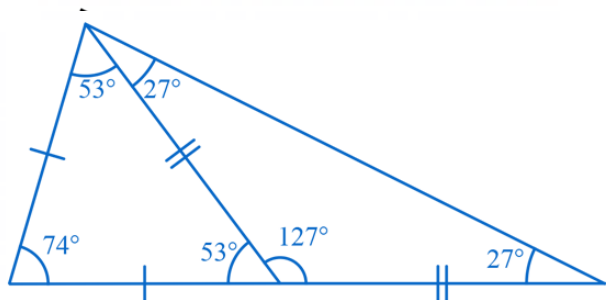
a



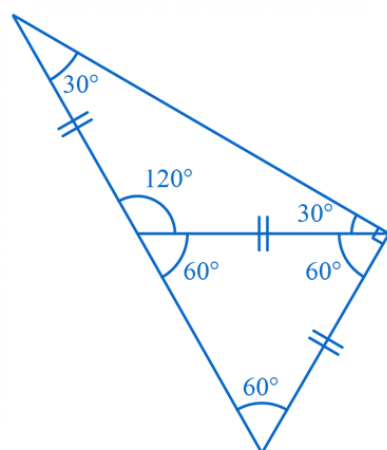
b



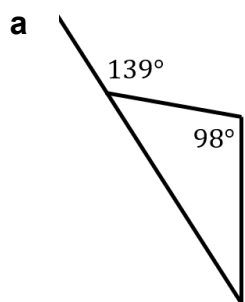
c



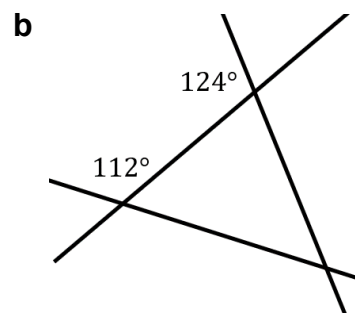
d



13 Decide whether the triangles shown are isosceles.



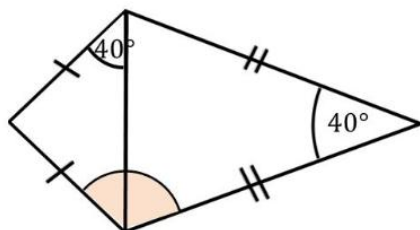
yes



yes

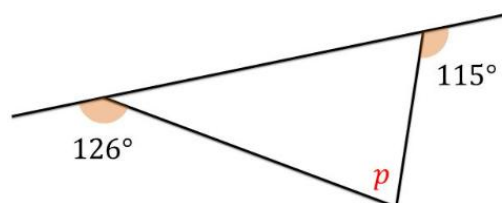
14

m Find the size of the shaded angle.



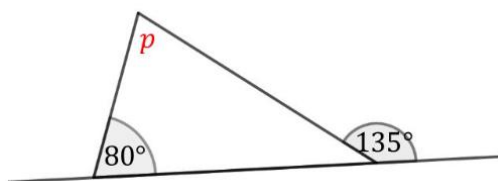
110°

n Find the size of angle  $p$ .



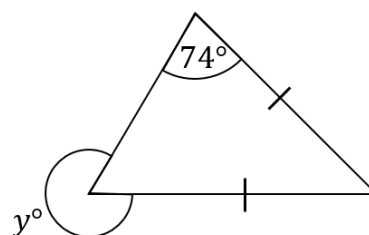
61°

o Find the size of angle  $p$ .



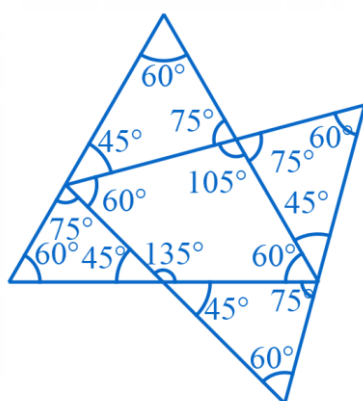
55°

p Find the size of angle  $y$ .

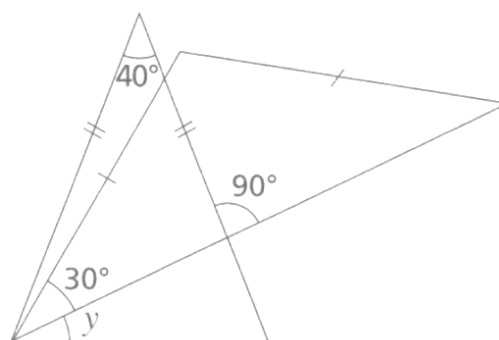


286°

q This is a pair of overlapping equilateral triangles. Find every angle.

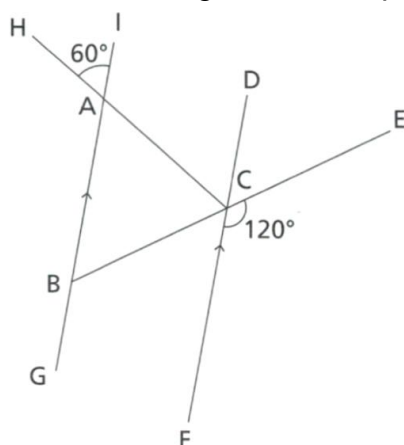


r The is a pair of isosceles triangles. Work out the size of angle  $y$ .



20°

15 Show that triangle ABC is equilateral. Give full reasoning for each stage of your working.



$$\angle BAC = 60^\circ \text{ (vertically opposite)}$$

$$\angle BCF = 60^\circ \text{ (angles on straight line)}$$

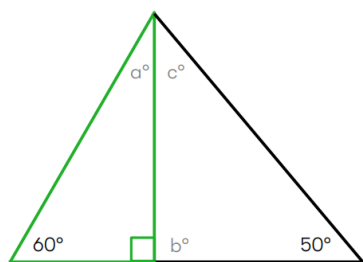
$$\begin{aligned} \angle ABC &= \angle BCF \text{ (alternate angles on parallel lines)} \\ &= 60^\circ \end{aligned}$$

$$\begin{aligned} \angle ACB &= 180^\circ - 60^\circ - 60^\circ \text{ (angle sum of } \triangle) \\ &= 60^\circ \end{aligned}$$

$$\therefore \triangle ABC \text{ is equilateral.}$$

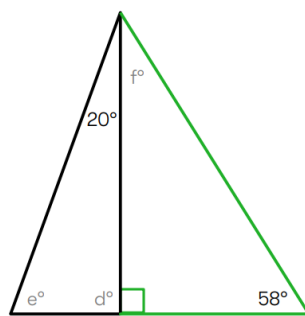
16 Determine the value of the pronumerals in these diagrams.

a



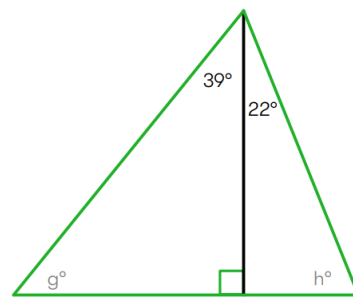
$$a = 30, b = 90, c = 40$$

b



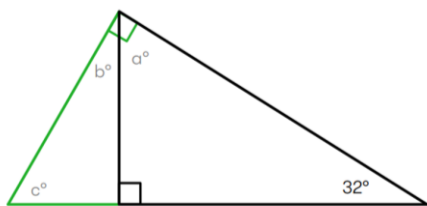
$$d = 90, e = 70, f = 32$$

c



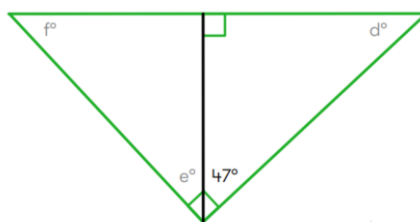
$$g = 51, h = 68$$

d



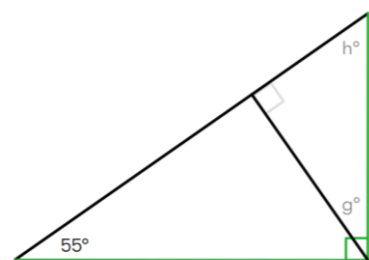
$$a = 58, b = 22, c = 58$$

e



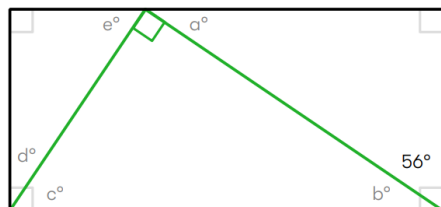
$$d = 43, e = 43, f = 47$$

f



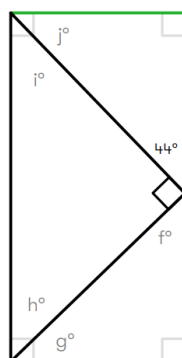
$$g = 55, h = 35$$

g



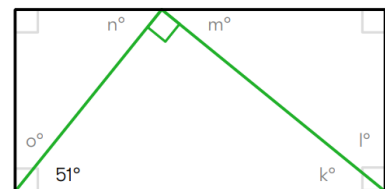
$$a = 34, b = 34, c = 56, d = 34, e = 56$$

h



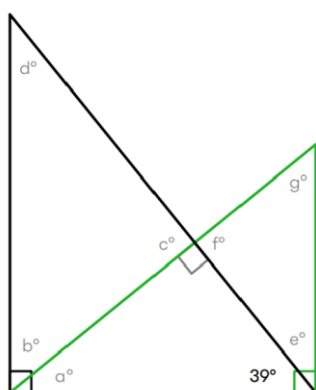
$$f = 46, g = 44, h = 46, i = 44, j = 46$$

i



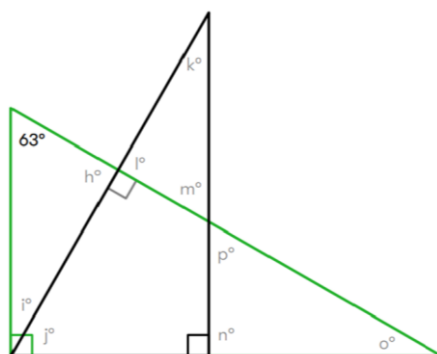
$$k = 39, l = 51, m = 39, n = 51, o = 39$$

j



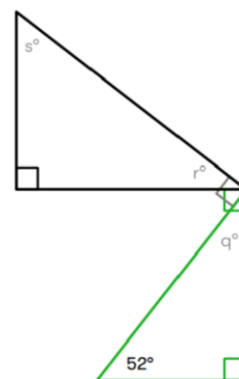
$$a = 51, b = 39, c = 90 \\ d = 51, e = 51, f = 90, g = 39$$

k



$$h = 90, i = 27, j = 63, k = 27, l = 90, \\ m = 63, n = 90, o = 27, p = 63$$

l

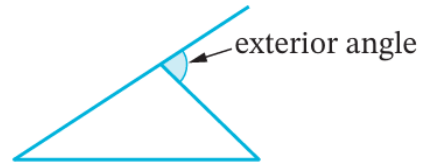
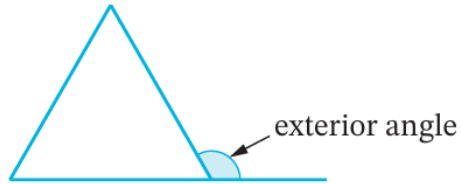


$$q = 38, r = 38, s = 52$$

# Exterior Angle Theorem

## i Exterior Angle of a Triangle

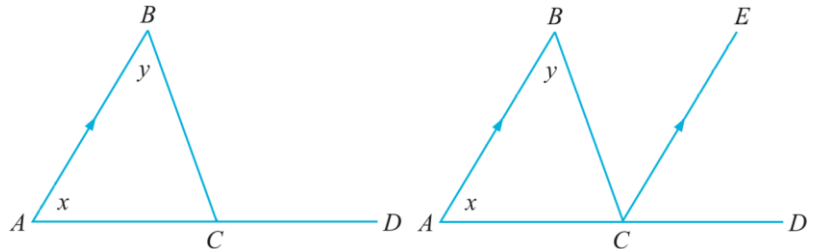
If we extend one side of a triangle, we make an **exterior angle**.



## Q Exterior Angle Theorem

Consider triangle  $\triangle ABC$  with exterior angle  $\angle BCD$ .

Construct line  $CE$  parallel to  $AB$



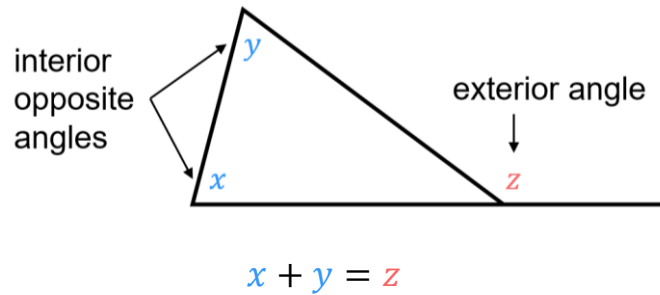
$$\angle BAC = \angle ECD \text{ (.....)}$$

$$\angle ABC = \angle \dots \text{ (alternate angles on } \parallel \text{ lines)}$$

$$\therefore \angle BCD = \dots + \dots$$

The exterior angle of a triangle is equal to the ..... of the two opposite interior angles.

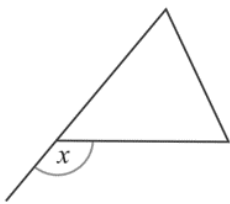
## i Exterior Angle Theorem



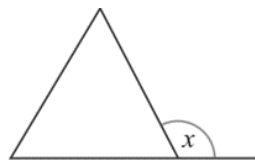
## Q Identify Exterior angles and interior opposite angles

Indicate the two interior angles opposite the exterior angle  $x$  in these diagrams

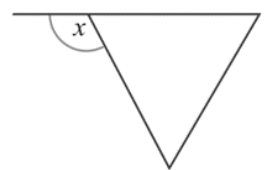
a



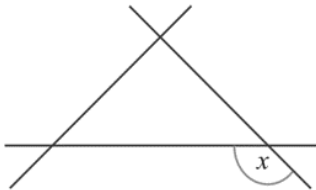
b



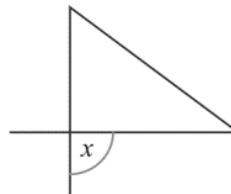
c



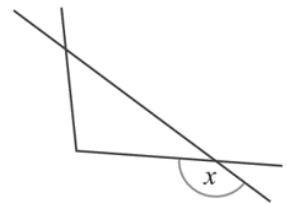
d



e

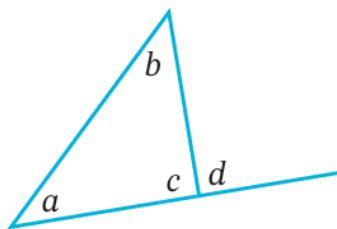


f



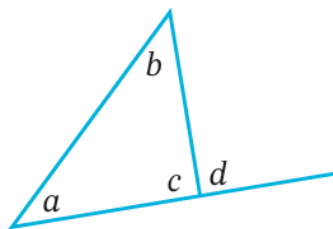
For each triangle, write an equation using exterior angle theorem.

a



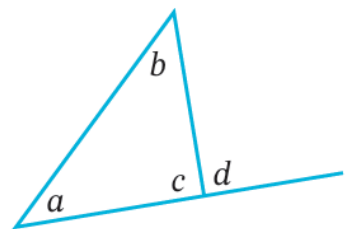
$$a + b = \dots\dots\dots$$

b



$$a + b = \dots\dots\dots$$

c

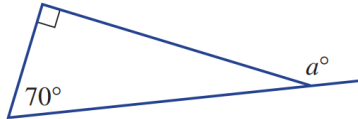


$$a + b = \dots\dots\dots$$

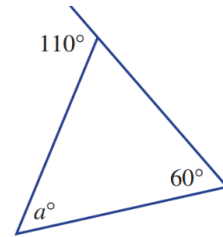
## 💡 Key Ideas

1. The exterior angle of a triangle is the ..... of the two interior opposite angles.

## **Example** Solve problems involving the exterior angle theorem



1.  $90 + 70 = a$  (exterior angle theorem)
3.  $160^\circ = a$

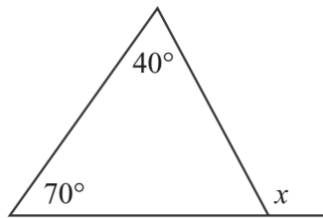


1.  $a + 60 = 110$  (exterior angle theorem)
3.  $a = 50^\circ$

## **Guided Practice**

Find the value of the pronumeral in these diagrams.

**a**

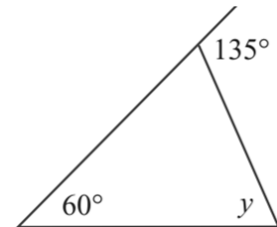


$$x = \dots + \dots$$

$$= \dots$$

$$x = 110^\circ$$

**b**

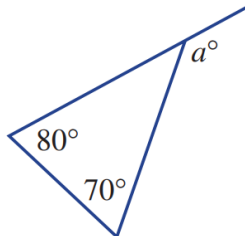


$$\dots + y = \dots$$

$$y = \dots$$

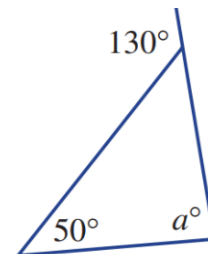
$$y = 75^\circ$$

**c**



$$a = 150^\circ$$

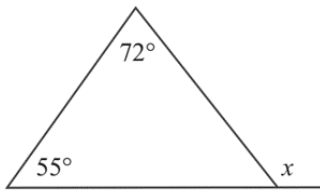
**d**



$$a = 80^\circ$$

1 Calculate the value of the missing exterior angle using exterior angle theorem.

a

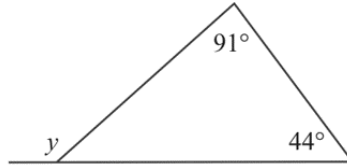


$$x = \dots + \dots$$

$$= \dots$$

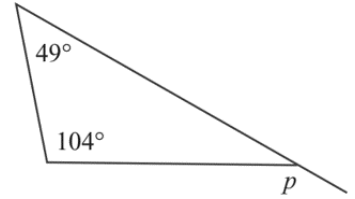
127°

b



135°

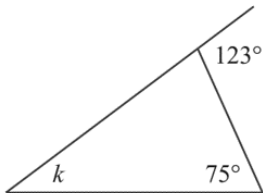
c



153°

2 Calculate the value of the missing interior angle using exterior angle theorem.

a

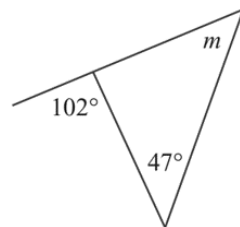


$$\dots = k + \dots$$

$$\dots = k$$

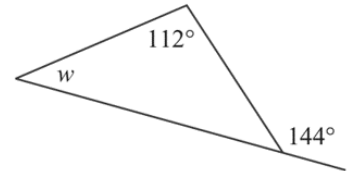
48°

b



55°

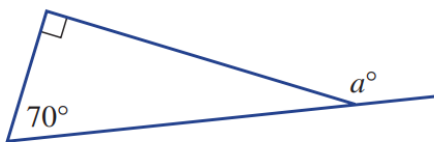
c



32°

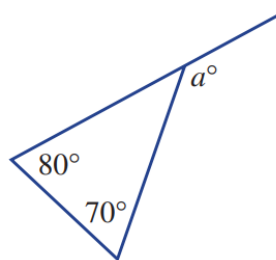
3 Calculate the value of the pronumeral. You do not need to show reasoning.

a



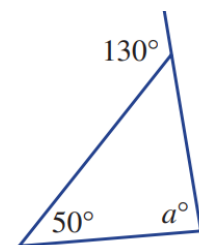
160°

b



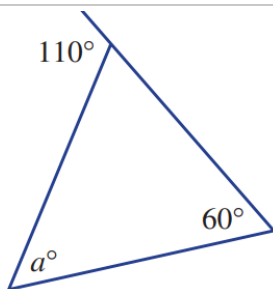
150°

c



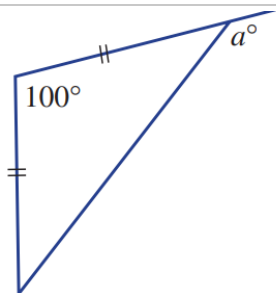
80°

d



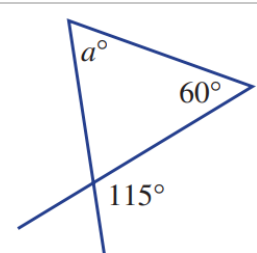
50°

e



140°

f



55°

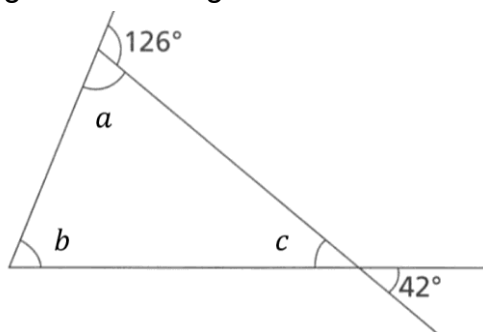
4 Find the value of each pronumeral, you do not need to show reasoning.

|                            |                                                     |                                                    |                                                     |
|----------------------------|-----------------------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| <p><b>a</b></p> <p>104</p> | <p><b>b</b></p> <p>16</p>                           | <p><b>c</b></p> <p><math>d = 85, e = 95</math></p> | <p><b>d</b></p> <p><math>a = 50, b = 116</math></p> |
| <p><b>e</b></p> <p>135</p> | <p><b>f</b></p> <p><math>x = 66, y = 114</math></p> | <p><b>g</b></p> <p>67</p>                          | <p><b>h</b></p> <p>29</p>                           |
| <p><b>i</b></p> <p>90</p>  | <p><b>j</b></p> <p>70</p>                           | <p><b>k</b></p> <p>130</p>                         | <p><b>l</b></p> <p>88</p>                           |

5 Determine the value of the pronumeral, using all the angle relationships you know. You do not need to show reasoning.

|                           |                             |                           |
|---------------------------|-----------------------------|---------------------------|
| <p><b>a</b></p> <p>60</p> | <p><b>b</b></p> <p>231</p>  | <p><b>c</b></p> <p>18</p> |
| <p><b>d</b></p> <p>91</p> | <p><b>e</b></p> <p>65.5</p> | <p><b>f</b></p> <p>60</p> |

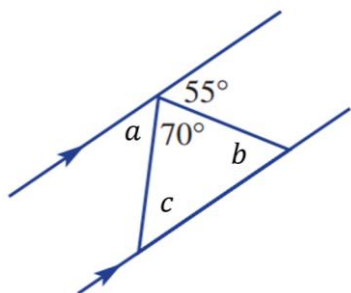
- 6 Determine all the missing angles in this diagram.



$$\begin{aligned} a &= 54^\circ \\ b &= 84^\circ \\ c &= 42^\circ \end{aligned}$$

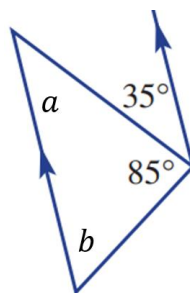
- 7 Use your knowledge of angle relationships on parallel lines to calculate the unknown angles.

a



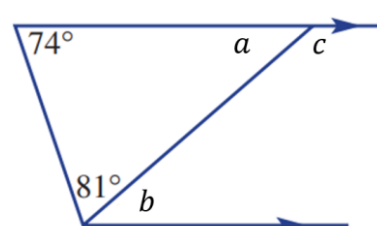
$$a = 55^\circ, b = 55^\circ, c = 55^\circ$$

b



$$a = 35^\circ, b = 60^\circ$$

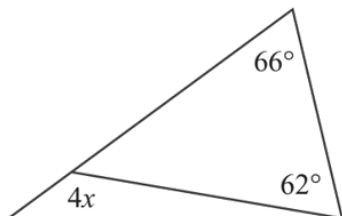
c



$$a = 25^\circ, b = 25^\circ, c = 155^\circ$$

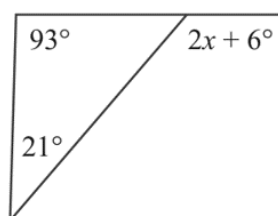
- 8 By writing an equation, solve for the pronumeral.

a



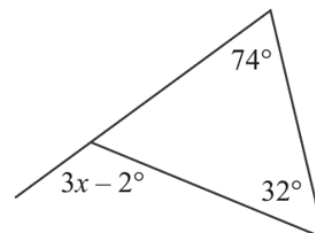
$$\begin{aligned} 4x + 52 &= 180 \\ 32^\circ \end{aligned}$$

b



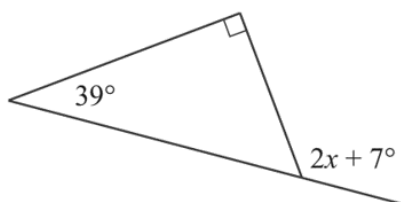
$$\begin{aligned} 2x + 72 &= 180 \\ 54^\circ \end{aligned}$$

c



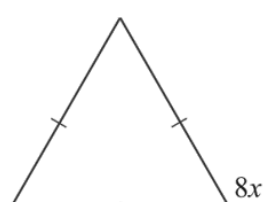
$$\begin{aligned} 3x + 72 &= 180 \\ 36^\circ \end{aligned}$$

d



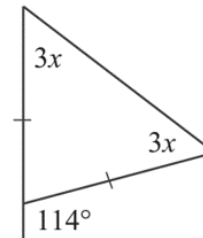
$$\begin{aligned} 2x + 7 &= 129 \\ 61^\circ \end{aligned}$$

e



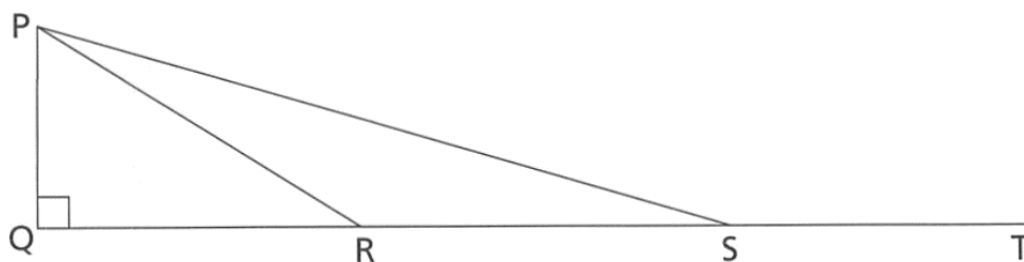
$$\begin{aligned} 8x + 60 &= 180 \\ 15^\circ \end{aligned}$$

f



$$\begin{aligned} 6x &= 114 \\ 14^\circ \end{aligned}$$

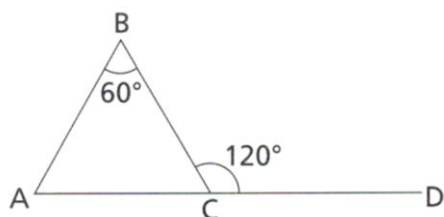
- 9  $\angle PRQ = 48^\circ$  and  $\angle TSP = 141^\circ$ . Determine the sizes of every angle.



$$\begin{aligned}\angle QPR &= 42^\circ \\ \angle PRS &= 132^\circ \\ \angle RSP &= 39^\circ \\ \angle RPS &= 9^\circ\end{aligned}$$

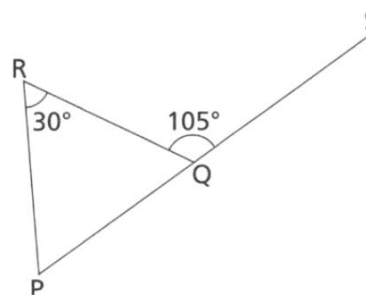
- 10 Show these properties, using full reasoning.

- a Show that triangle  $ABC$  is equilateral.



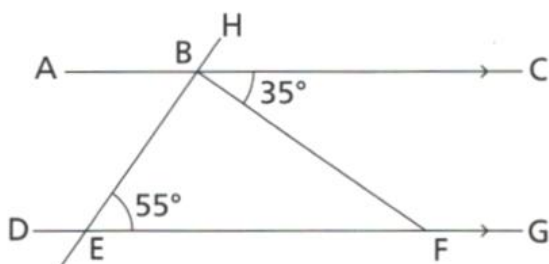
$$\begin{aligned}\angle BAC &= 120 - 60 \text{ (exterior angle theorem)} \\ &= 60^\circ \\ \angle BCA &= 180 - 120 \text{ (angles on a straight line)} \\ &= 60^\circ \\ \therefore \triangle ABC &\text{ is equilateral.}\end{aligned}$$

- b Show that triangle  $PQR$  is isosceles.



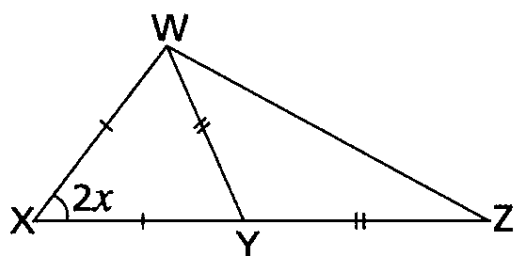
$$\begin{aligned}\angle RPQ &= 105 - 30 \text{ (exterior angle theorem)} \\ &= 75^\circ \\ \angle RQP &= 180 - 105 \text{ (angles on a straight line)} \\ &= 75^\circ \\ \angle RPQ &= \angle RQP \\ \therefore \triangle PQR &\text{ is isosceles.}\end{aligned}$$

- 11 Prove that  $\triangle BEF$  is a right-angled triangle, giving full geometrical reasoning.



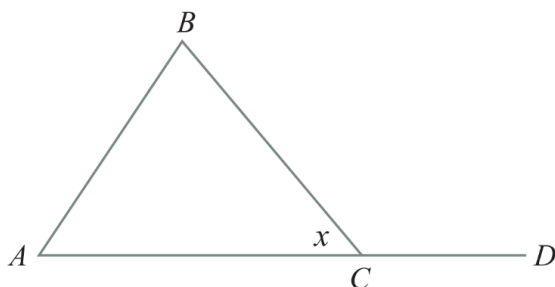
$$\begin{aligned}\angle CBF &= \angle BFE = 35^\circ \text{ (alternate angles on parallel lines)} \\ \angle EBF &= 180 - 55 - 35 \\ &= 90^\circ \text{ (angle sum of triangle)} \\ \therefore \triangle BEF &\text{ is right-angled}\end{aligned}$$

- 12 Find expressions, in terms of  $x$ , for all the angles in this triangle.  
(Hint: simplify the expressions as you go)



$$\begin{aligned}\angle XWY &= \angle XYW = 90 - x \\ \angle WYZ &= 90 + x \\ \angle YWZ &= \angle WZY = 45 - \frac{x}{2}\end{aligned}$$

- 13 Another proof of the alternate angle theorem is possible by using the angle sum of a triangle.



By considering  $\triangle ABC$ ,  $x + \angle BAC + \angle ABC = 180$  (.....)

Angle sum of  $\triangle$

$$\angle BAC + \angle ABC = 180 - \dots\dots\dots$$

By considering  $\angle ACD$ ,  $x + \angle BCD = \dots\dots\dots$  (.....)

$x$

180 (angles on straight lines)

$$\angle BCD = \dots\dots\dots$$

$180 - x$

To summarise:

$$\angle BAC + \angle ABC = 180 - x$$

$$\angle BCD = 180 - x$$

So  $\dots\dots\dots = \dots\dots\dots$

$$\angle BAC + \angle ABC = \angle BCD$$

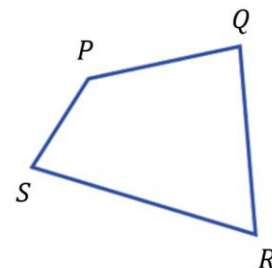
# Features of Quadrilaterals

## i Labelling Quadrilaterals

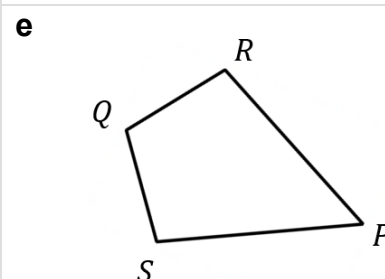
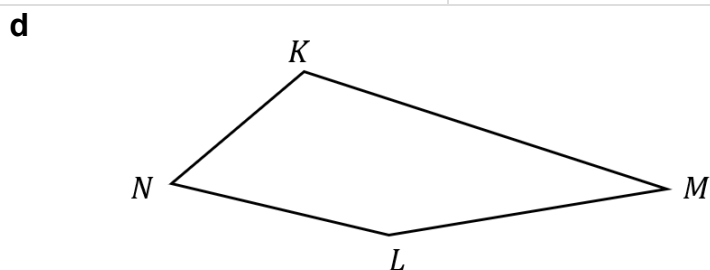
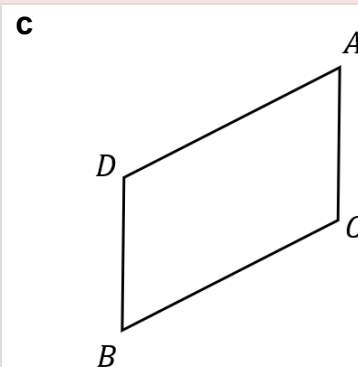
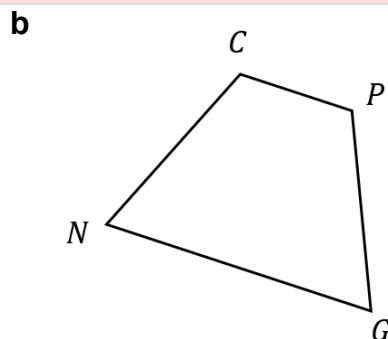
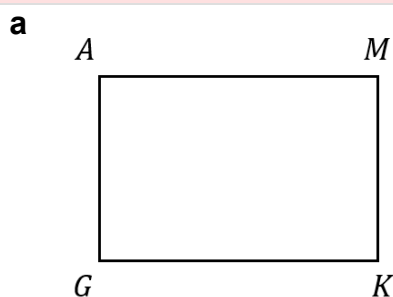
- A **quadrilateral** is a **4-sided** shape.
- We label quadrilaterals using their vertices.
- Write vertices in clockwise or anticlockwise order.

$PQRS$  or  $PSRQ$  ✓

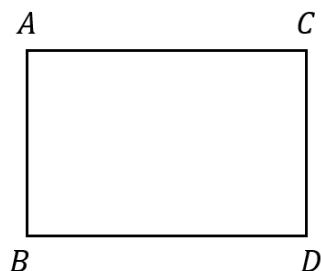
$PRQS$  ✗



## Label quadrilaterals

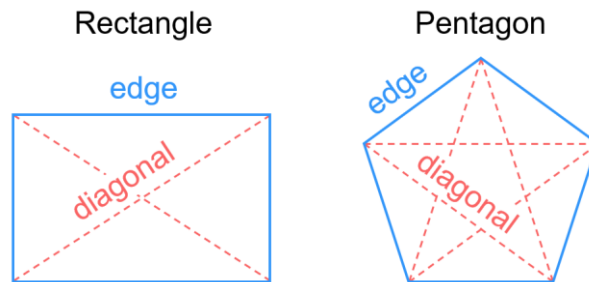


- f** Tommy was asked to draw a rectangle  $ABCD$ . This is what he sketched. Explain why he is incorrect, and draw rectangle  $ABCD$  correctly.

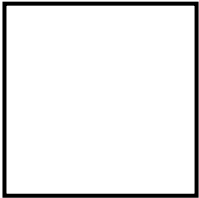
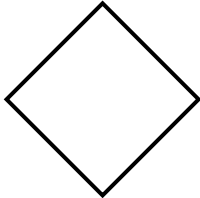
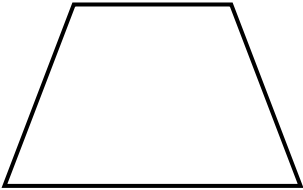
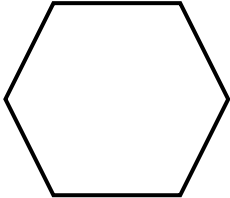
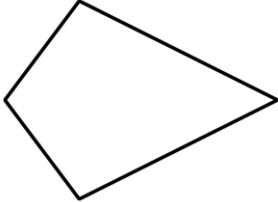
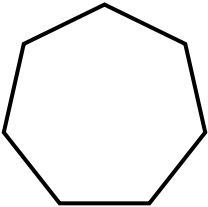
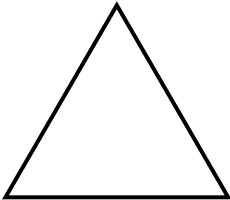
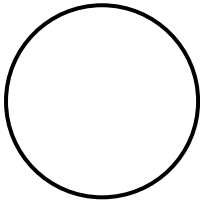


## i Diagonals

- **Adjacent** means next to.
- An **edge** of a polygon joins two **adjacent** vertices.
- A **diagonal** of a polygon joins two **non-adjacent vertices**.
- A diagonal goes from one corner to another, skipping at least one corner.

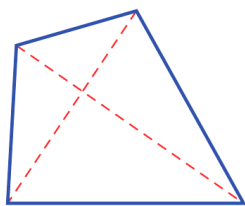


## Q Draw Diagonals of a polygon

|                                                                                                 |                                                                                                 |                                                                                                   |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>a</b><br>  | <b>b</b><br>  | <b>c</b><br>  |
| <b>d</b><br> | <b>e</b><br> | <b>f</b><br> |
| <b>g</b><br> | <b>h</b><br> | <b>i</b><br> |

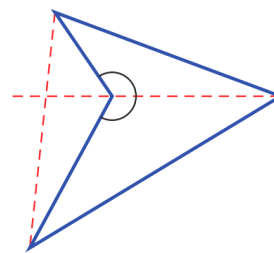
## i Convex and Non-Convex Quadrilaterals

### Convex



All interior angles less than  $180^\circ$   
Diagonals meet inside the quadrilateral.

### Non-Convex

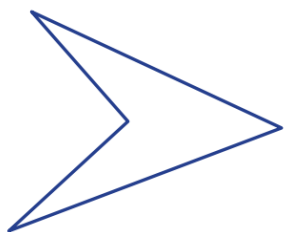


At least one interior reflex angle ( $>180^\circ$ )  
Diagonals meet outside the quadrilateral.

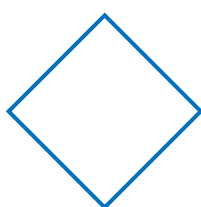
## Q Identify Convex and Non-Convex Quadrilaterals

Determine whether these quadrilaterals are convex or non-convex by drawing the diagonals.

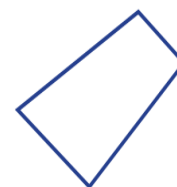
a



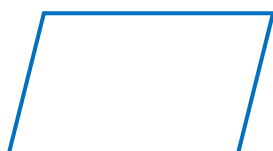
b



c



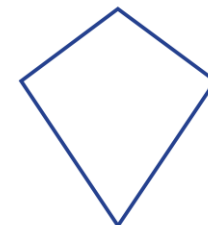
d



e



f



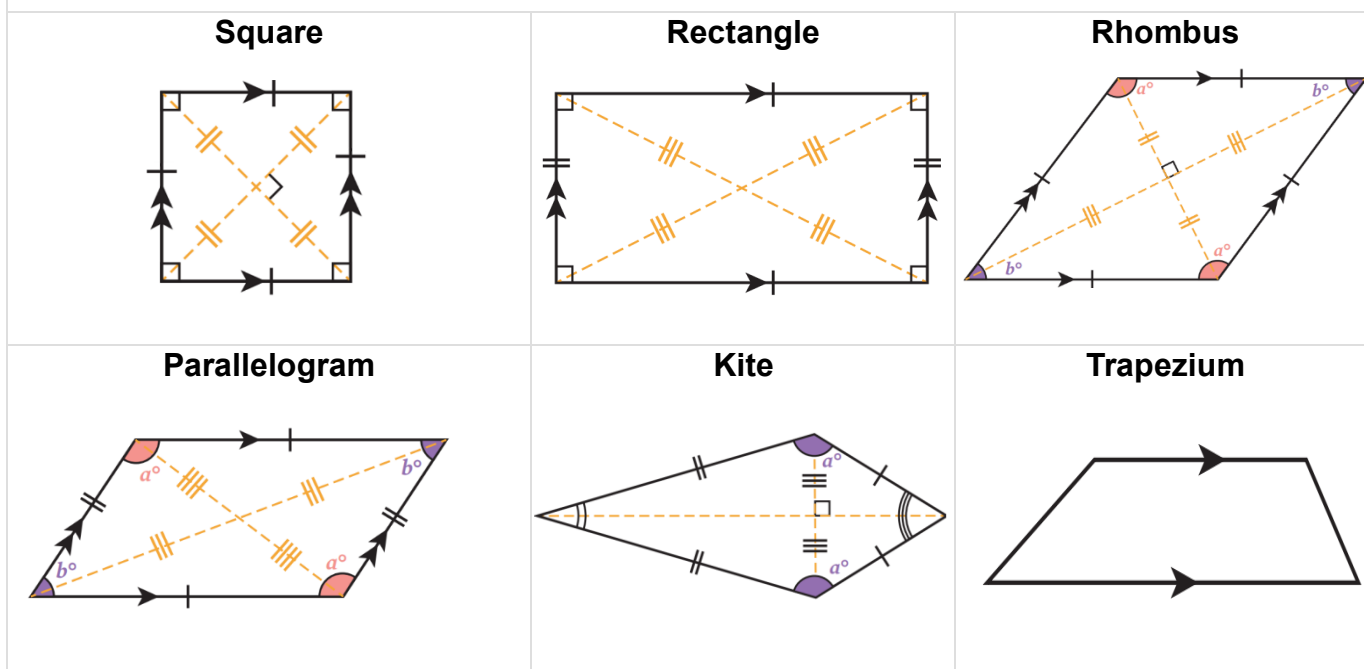
# Properties of Special Quadrilaterals

## Q Properties of Special Quadrilaterals

**Adjacent** means next to. **Bisect** means cut into two equal parts.

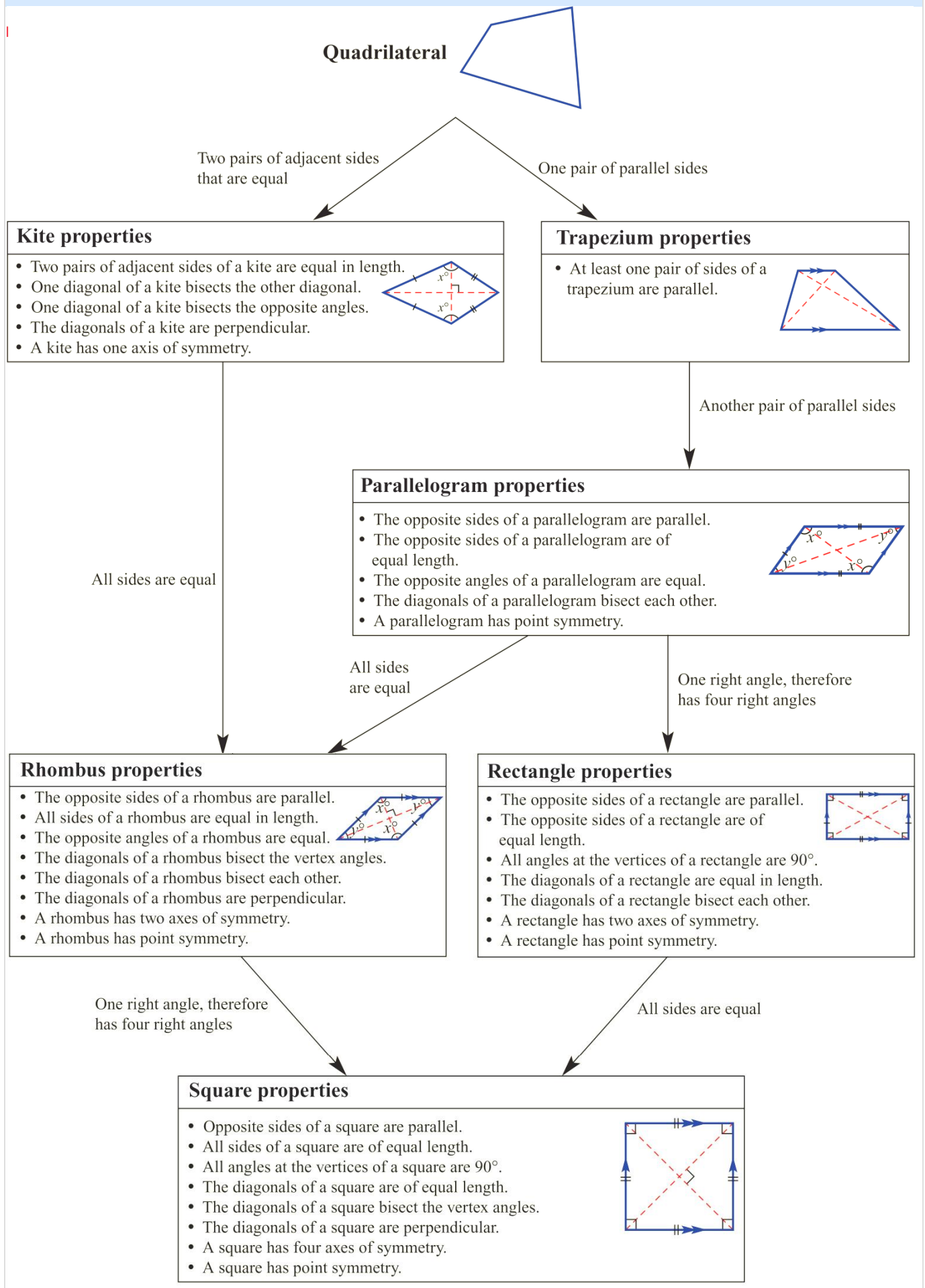
Using the notation on the diagrams, or the GeoGebra links, fill in the table.

- 🔍 Properties of a Square: <https://www.geogebra.org/m/xq7pbnpe>
- 🔍 Properties of a Rectangle: <https://www.geogebra.org/m/axwkzwq6>
- 🔍 Properties of a Rhombus: <https://www.geogebra.org/m/wtu2b73s>
- 🔍 Properties of a Parallelogram: <https://www.geogebra.org/m/gmhdqgpn>
- 🔍 Properties of a Parallelogram: <https://www.geogebra.org/m/qem8ddkv>
- 🔍 Properties of a Trapezium: <https://www.geogebra.org/m/h2kxb78k>



| Property (always true?)                      | Square | Rectangle | Rhombus | Parallelogram | Kite    | Trapezium |
|----------------------------------------------|--------|-----------|---------|---------------|---------|-----------|
| Opposite sides parallel                      | Yes    | Y         | Y       | Y             | No      | 1 pair    |
| Adjacent sides equal                         | Yes    | Y         | Y       | N             | 2 pairs | N         |
| Opposite sides equal                         | Yes    | Y         | Y       | Y             | No      | N         |
| Opposite angles equal                        | Yes    | Y         | Y       | Y             | 1 pair  | N         |
| Adjacent sides perpendicular                 | Yes    | Y         | Y       | N             | No      | N         |
| Diagonals bisect the other diagonal          | Yes    | Y         | Y       | Y             | 1 pair  | N         |
| Diagonals intersect at right angles          | Yes    | N         | Y       | N             | Yes     | N         |
| Diagonals equal                              | Yes    | Y         | N       | N             | No      | N         |
| Diagonals bisect angles of the quadrilateral | Yes    | N         | Y       | N             | 1 pair  | N         |

## *i* Hierarchy of Quadrilaterals



- 1 Identify the quadrilateral using its angle properties.  
Circle ALL of the shapes the quadrilaterals could be classified as.

**a**

10 cm 17 cm 10 cm 17 cm

Quadrilateral Kite  
Trapezium Rhombus  
Parallelogram Square  
Rectangle

Quadrilateral kite

**b**

Quadrilateral Kite  
Trapezium Rhombus  
Parallelogram Square  
Rectangle

Quadrilateral, parallelogram, rectangle

**c**

16 cm 4.2 cm 16.6 cm 9 cm

70° 133° 65° 92°

Quadrilateral Kite  
Trapezium Rhombus  
Parallelogram Square  
Rectangle

quadrilateral

**d**

124° 110° 56° 70°

Quadrilateral Kite  
Trapezium Rhombus  
Parallelogram Square  
Rectangle

Quadrilateral, trapezium

- 2 Name these special quadrilaterals using the most appropriate definition.

**a**

3 cm 3 cm 3 cm 3 cm

square

**b**

rectangle

**c**

trapezium

**d**

8 m 5 m 8 m 5 m

kite

**e**

121° 59° 121° 59°

parallelogram

**f**

parallelogram

**g**

rectangle

**h**

21 m 153° 21 m 27° 21 m 153° 21 m

rhombus

- 3 Explain why a square is a special type of rectangle.

A square has all the properties of a rectangle

- 4 Explain why a rectangle is not a type of square.

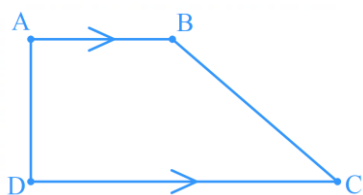
A rectangle does not have all the properties of a square; it does not have equal sides.

- 5 By considering the properties of special quadrilaterals, decide if the following are **always** true.

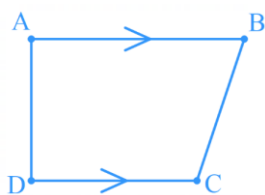
- a** A square is a type of rhombus. .... true
- b** A rectangle is a type of parallelogram. .... true
- c** A parallelogram is a type of square. .... false
- d** A rhombus is a type of parallelogram. .... true
- e** A parallelogram is a type of rhombus. .... false
- f** A rhombus is a type of kite. .... true

6 Sketch a:

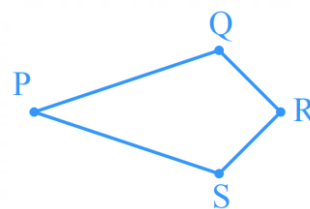
a trapezium ABCD, with AB parallel to CD



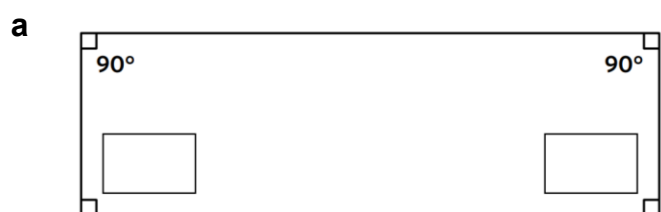
b trapezium ABCD, with  $\angle ABC$  acute



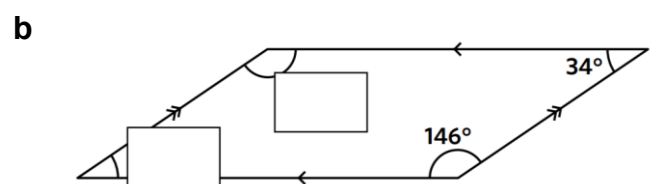
c kite, PQRS, with  $\angle PQR$  and  $\angle PSR$  both obtuse



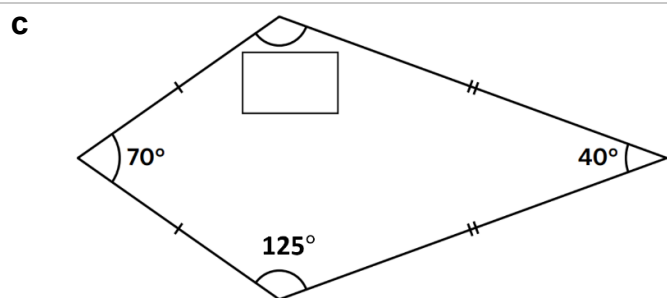
7 Find the missing angles. You will need to use properties of quadrilaterals.



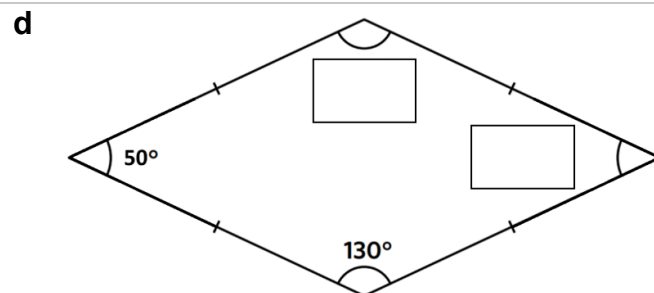
90°, 90°



34°, 146°



125°



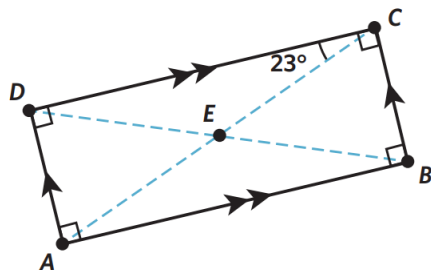
130°, 50°

8 First identify the quadrilateral, then determine the values of the angles.

a Quadrilateral:

.....

$\angle ECB =$  .....

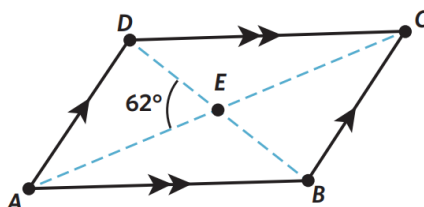


Rectangle  
67°

b Quadrilateral:

.....

$\angle CEB =$  .....

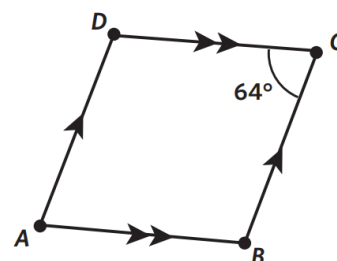


Parallelogram  
62°

c Quadrilateral:

.....

$\angle ABC =$  .....

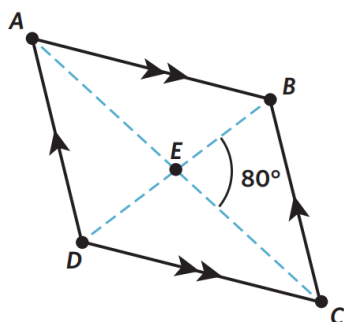


Parallelogram  
116°

d Quadrilateral:

.....

e  $\angle CED =$  .....

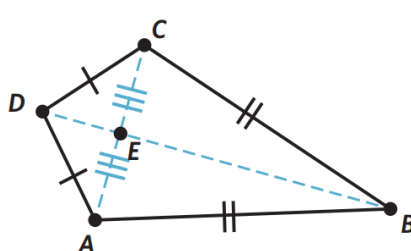


Parallelogram  
100°

f Quadrilateral:

.....

g  $\angle DEA =$  .....

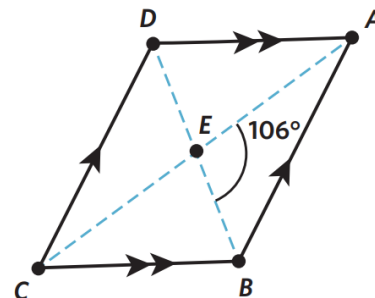


Kite  
90°

h Quadrilateral:

.....

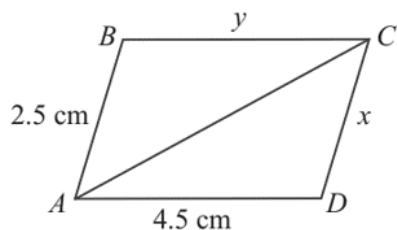
i  $\angle ECD + \angle EDC =$  .....



Parallelogram  
74°

9 Use the properties of quadrilaterals to find the values of the pronumerals, giving reasons.

a  $ABCD$  is a parallelogram.

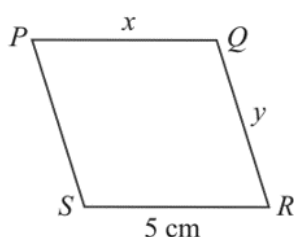


$x = 2.5$  cm (opposite sides of a parallelogram equal)

$y = \dots\dots\dots$  (.....)

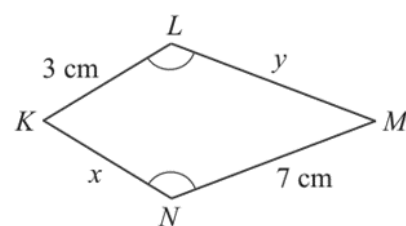
4.5 cm (opposite sides of parallelogram equal)

b  $PQRS$  is a rhombus.



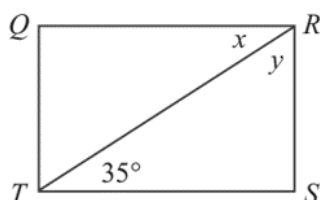
$x = y = 5$  cm  
All sides of rhombus equal

c  $KLMN$  is a kite.



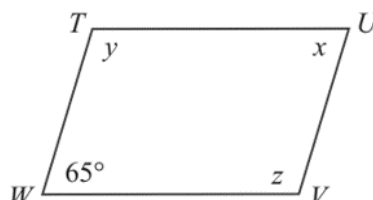
$x = 3$   
 $y = 7$   
Adjacent sides of kite are equal

d  $QRST$  is a rectangle.



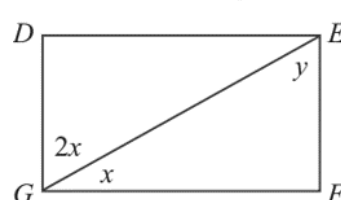
$x = 35^\circ$  Alternate angles on parallel lines  
 $y = 55^\circ$  complementary with  $x$

e  $TUVW$  is a parallelogram.



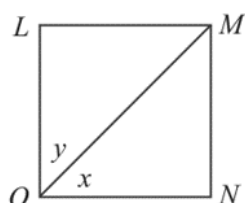
$y = 115^\circ$  Cointerior angles in parallel lines  
 $z = 115^\circ$  Cointerior angles in parallel lines  
 $x = 65^\circ$  opposite angles in parallelogram equal

f  $DEFG$  is a rectangle.



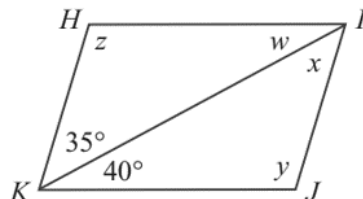
$3x = 90$  angles in a rectangle  
 $x = 30^\circ$   
 $y = 2x$  alternate angles on parallel lines  
 $x = 60^\circ$

g  $LMNO$  is a square.



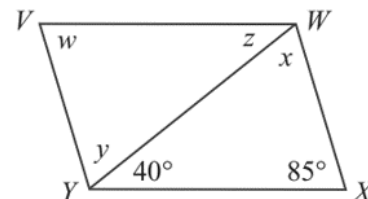
$x = y$  diagonals of square bisect angles at  
 $x + y = 90^\circ$  (angles in a square)  
 $x = y = 45^\circ$

h  $HIJK$  is a parallelogram.



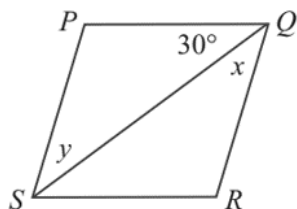
$x = 35^\circ$ ,  $w = 40^\circ$   
alternate angles on parallel lines  
 $z = 105^\circ$  cointerior angles in parallel lines  
 $y = z = 105^\circ$   
opposite angles of parallelogram equal

i  $VWXY$  is a parallelogram.



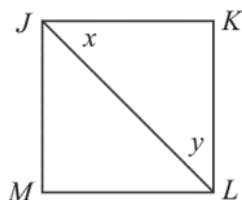
$z = 40^\circ$  alternate angles on parallel lines  
 $y = 55^\circ$  cointerior angles on parallel lines  
 $x = y = 55^\circ$  alternate angles on parallel lines  
 $w = 85^\circ$  opposite angles of parallelogram

**j**  $PQRS$  is a rhombus.



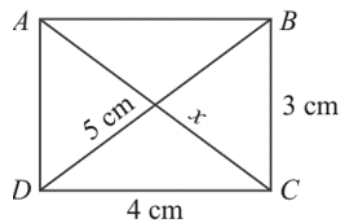
$x = 30^\circ$   
 diagonals in rhombus bisect angles at vertex  
 $y = x = 30^\circ$  alternate angles on parallel lines

**k**  $JKLM$  is a square.



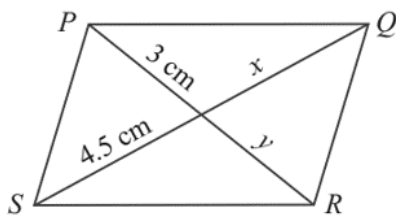
$x = y = 45^\circ$   
 diagonals in square bisect angles at vertex

**l**  $ABCD$  is a rectangle.



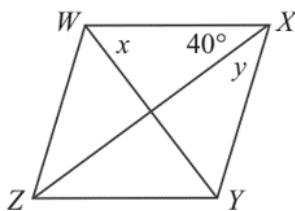
$x = 5$  cm diagonals of rectangle are equal and bisect each other.

**m**  $PQRS$  is a parallelogram.



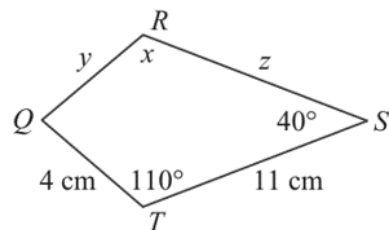
$x = 4.5$  cm,  $y = 3$  cm  
 Diagonals of parallelogram bisect each other

**n**  $WXYZ$  is a rhombus.



$y = 40^\circ$  diagonals of rhombus bisect angles at vertex  
 $x = 50^\circ$  angle sum of triangle

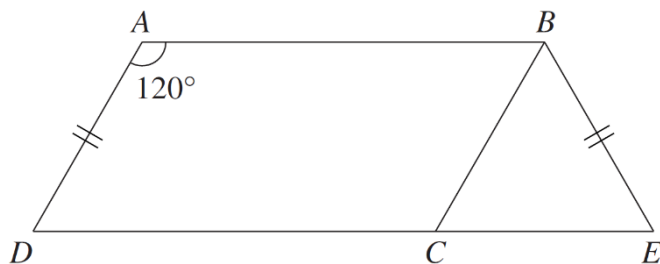
**o**  $QRST$  is a kite.



$x = 110^\circ$  opposite angles of kite between unequal sides equal.  
 $y = 4$  cm,  $x = 11$  cm, adjacent sides of kite equal

# 10 NAPLAN B+

The diagram shows a parallelogram  $ABCD$ . Prove that  $\triangle BCE$  is equilateral.



$\angle BCD = 120^\circ$  opposite angles of parallelogram equal.

$\angle ECB = 60^\circ$  angles on a straight line

$BC = AD$  opposite sides of parallelogram equal.

$\therefore BC = AD = BE$ , so  $\triangle BCE$  is isosceles

$\therefore \angle BCE = \angle BEC = 60^\circ$  angles opposite equal sides of isosceles triangle are equal

$\angle BCE = 180 - 60 - 60 = 60^\circ$  angle sum of triangle

$\therefore \triangle BCE$  is equilateral, as all angles are  $60^\circ$

- 11 The length of the diagonals of a rhombus is 24 cm and 18 cm.  
Find the length of each side of the rhombus.  
(Hint: use Pythagoras' Theorem)

The diagonals form 4 right-angled triangles where the perpendicular sides are 12 cm and 9 cm.

The hypotenuse is the side length of the rhombus.

$$l = \sqrt{12^2 + 9^2} = 15 \text{ cm}$$

- 12 The side of a rhombus is 5 cm.  
The length of one diagonal of the rhombus is 8 cm.  
Find the length of the other diagonal.

The diagonals form 4 right-angled triangles where the hypotenuse is 5 cm and one of the shorter sides is 4 cm.

The other diagonal is double the other side length.

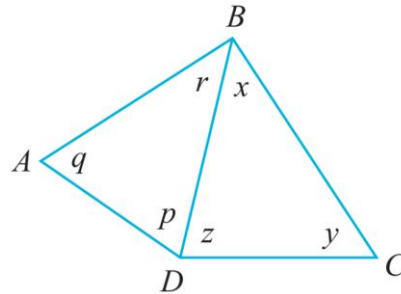
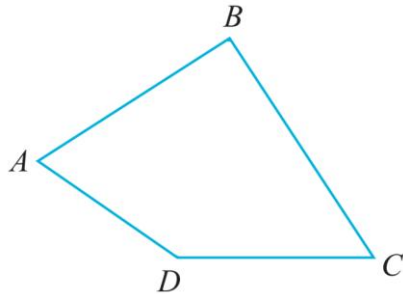
$$l = \sqrt{5^2 - 4^2} = 3$$

6 cm

# Angle Sum of a Quadrilateral

## Q Angle Sum of a Quadrilateral

Every quadrilateral can be split into two triangles.



By considering  $\triangle ABD$ ,  $q + p + r = 180^\circ$  (.....)

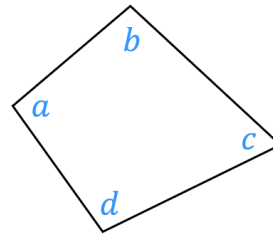
By considering  $\triangle BCD$ ,  $x + y + z = \dots\dots\dots$  (.....)

So, the sum of all angles in a quadrilateral is

$$\begin{aligned} & (q + p + r) + (x + y + z) \\ &= \dots\dots\dots + \dots\dots\dots \\ &= \dots\dots\dots \end{aligned}$$

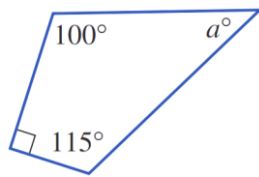
## i Angle Sum of a Quadrilateral

$$a + b + c + d = 360^\circ$$

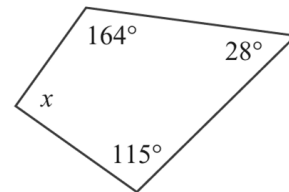


1. Write an equation using angle relationships.
2. Simplify the equation, if needed.
3. Solve for the unknown using inverse operations.

## Example Solve problems involving the angle sum of a quadrilateral



1.  $a + 90 + 100 + 115 = 360$  (angle sum of quad)
2.  $a + 305 = 360$
3.  $a = 55$

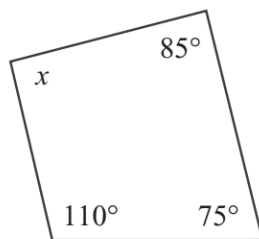


1.  $x + 164 + 115 + 28 = 360$  (angle sum of quad)
2.  $x + 307 = 360$
3.  $x = 53$

## Guided Practice

Find the value of the pronumerals in these quadrilaterals.

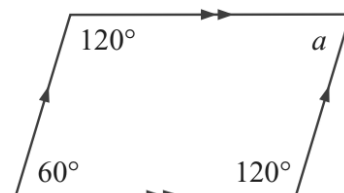
a



$$\begin{aligned} x + \dots + \dots + \dots &= \dots \\ x + \dots &= \dots \\ x &= \dots \end{aligned}$$

$$x = 90^\circ$$

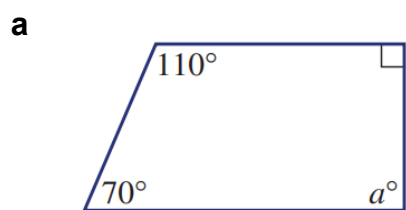
b



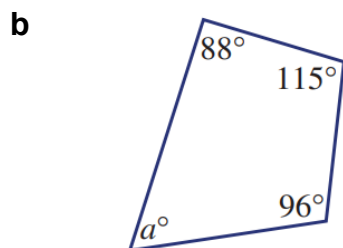
$$a + \dots + \dots + \dots = \dots$$

$$a = 60^\circ$$

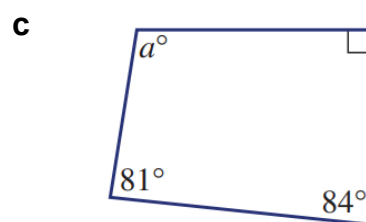
1 Determine the size of each unknown angle in these quadrilaterals.



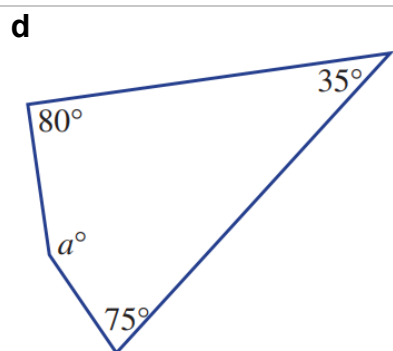
90



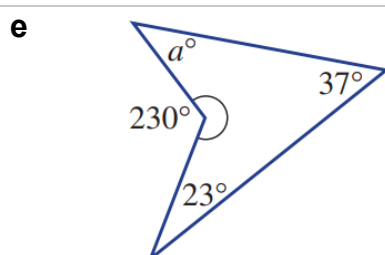
61



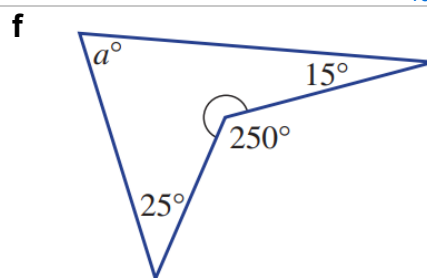
105



170

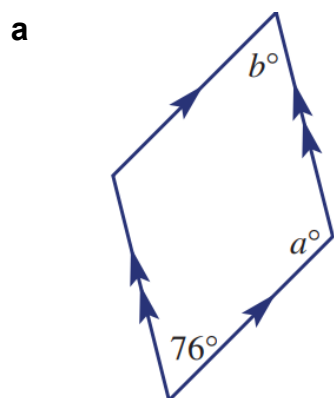


70

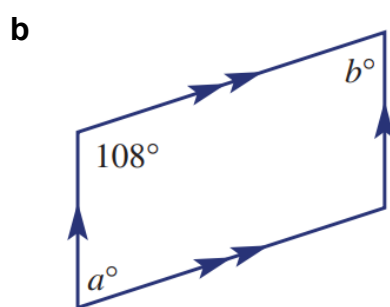


70

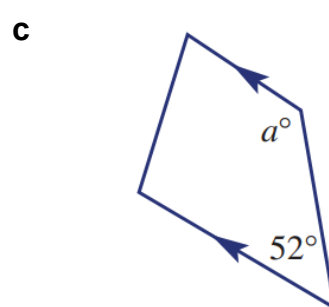
2 Using your knowledge of co-interior angles, determine the value of these pronumerals.



$a = 104, b = 76$

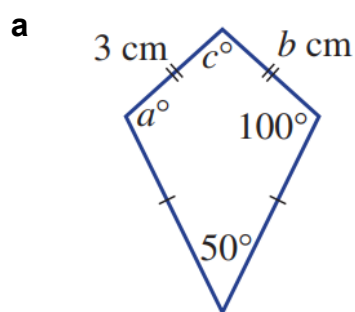


$a = 72, b = 72$

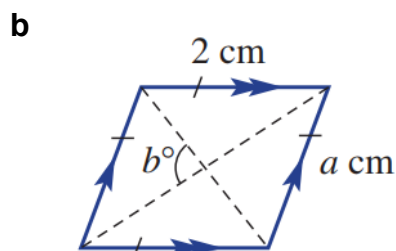


$a = 128$

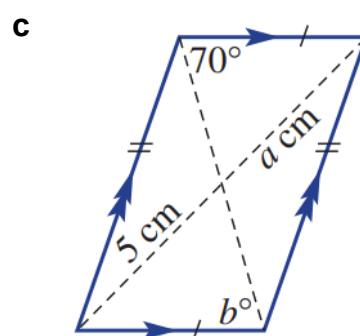
3 Using properties of special quadrilaterals, determine the value of these pronumerals.



$a = 100, b = 3, c = 110$



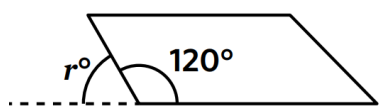
$a = 2, b = 90$



$a = 5, b = 70$

4 Find the value represented by the pronumeral.

a



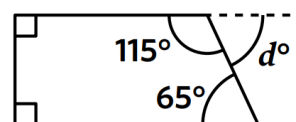
60

b



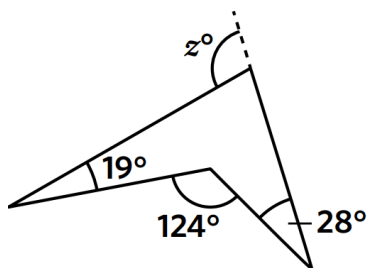
90

c



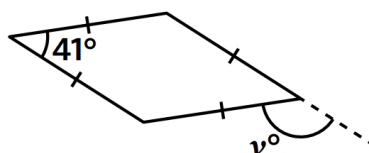
65

d



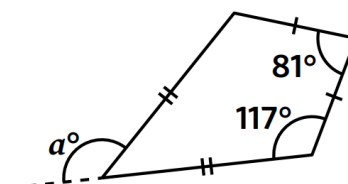
103

e



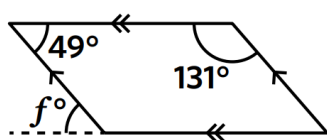
139

f



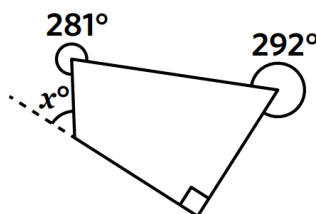
135

g



49

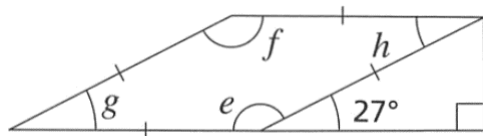
h



57

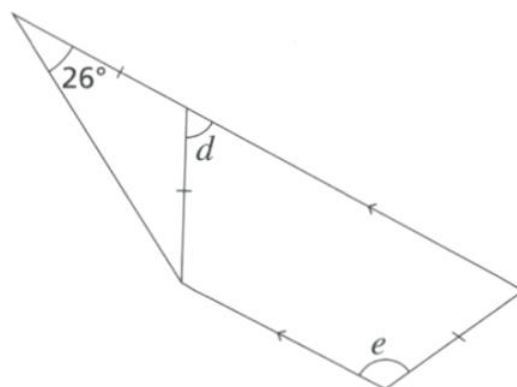
5 Find the values of the pronumerals.

a



$$e = 153, f = 153, g = 27, h = 27$$

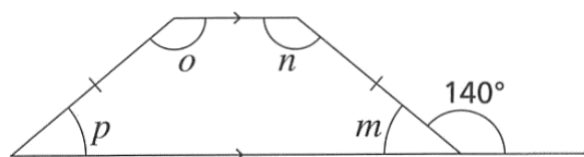
b



$$d = 52, e = 128$$

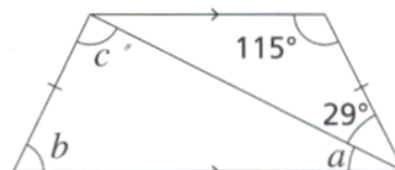
6 These two questions involve **isosceles trapeziums**. Adjacent angles in isosceles trapeziums are equal. Work out the size of each angle labelled with a letter.

a



$$m = p = 40^\circ \\ o = n = 140^\circ$$

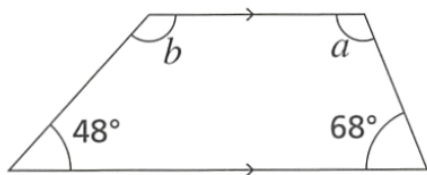
b



$$a = 36, b = 65, c = 79$$

7 Work out the values of the letters in these diagrams.

a



$$a = 112^\circ$$

$$b = 132^\circ$$

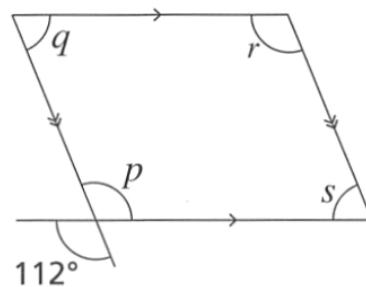
b



$$x = 125^\circ$$

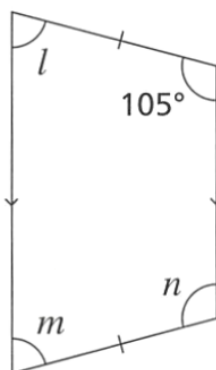
$$y = 64^\circ$$

c



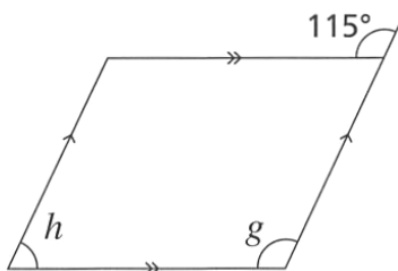
$$p = 112^\circ, q = 68^\circ, r = 112^\circ, s = 68^\circ$$

d



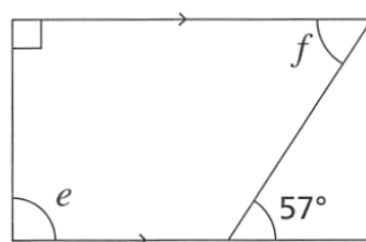
$$l = 75^\circ, m = 75^\circ, n = 105^\circ$$

e



$$g = 115^\circ, h = 65^\circ$$

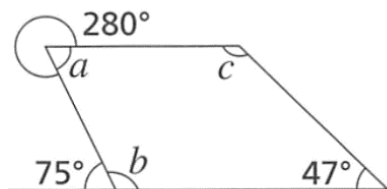
f



$$e = 90^\circ, f = 57^\circ$$

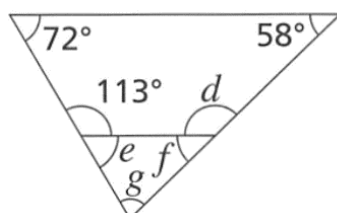
8 Determine the value of these pronumerals using your angle relationships knowledge.

a



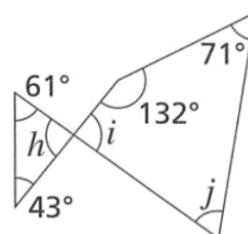
$$a = 80^\circ, b = 105^\circ, c = 128^\circ$$

b



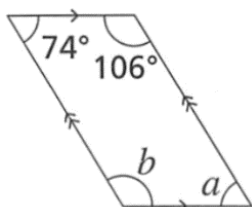
$$d = 117^\circ, e = 67^\circ, f = 63^\circ, g = 50^\circ$$

c



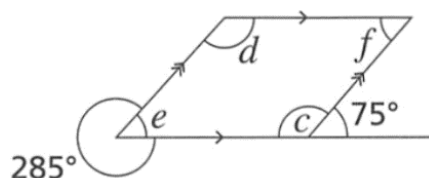
$$h = i = 76^\circ, j = 81^\circ$$

d



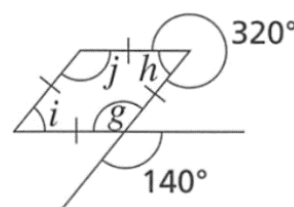
$$a = 74^\circ, b = 106^\circ$$

e



$$c = d = 105^\circ, e = f = 75^\circ$$

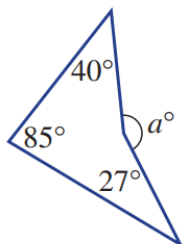
f



$$i = h = 40^\circ, j = g = 140^\circ$$

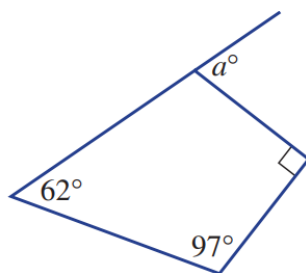
9 Determine the value of these pronumerals using your angle relationships knowledge.

a



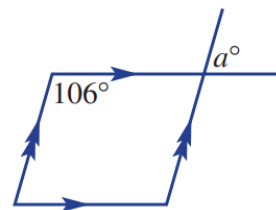
152

b



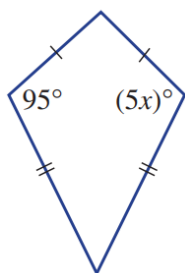
69

c



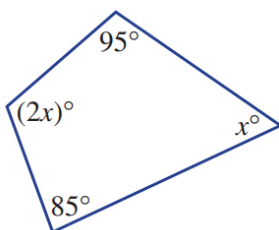
74

d



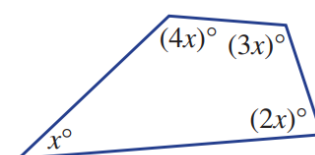
19

e



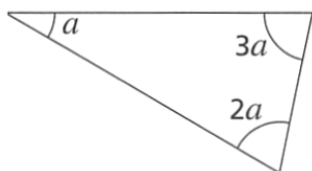
60

f



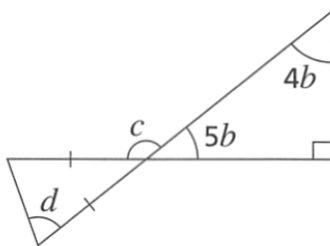
36

g



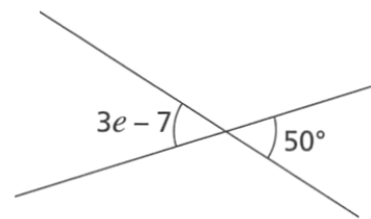
30

h



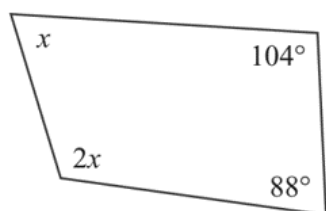
$b = 10, c = 130, d = 65$

i



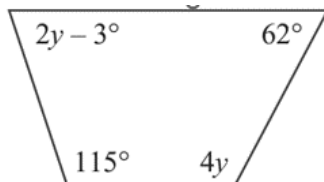
19

j



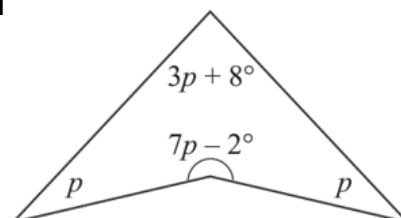
56

k



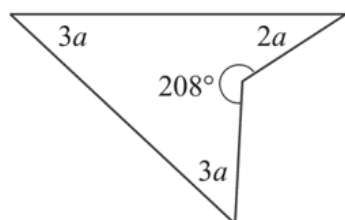
31

l



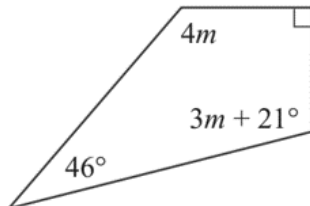
29.5

m



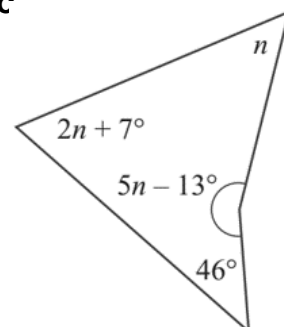
19

n



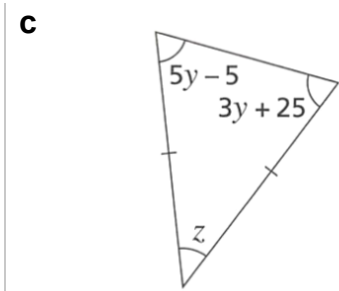
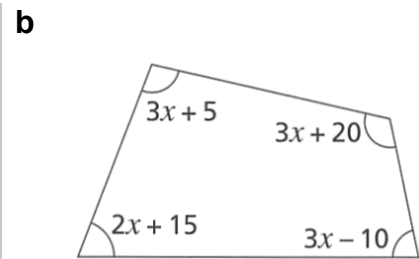
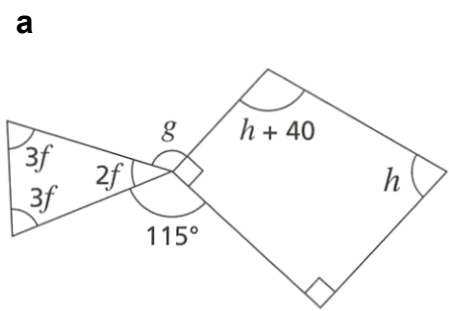
29

c



40

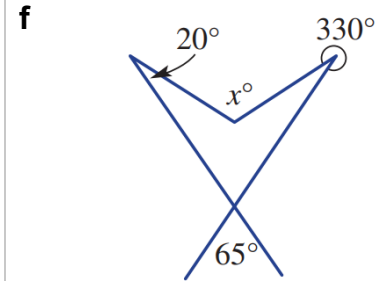
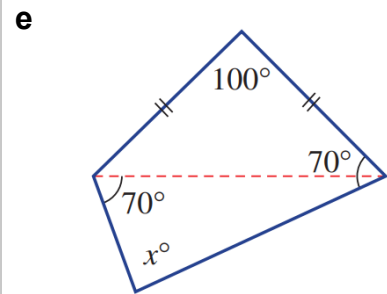
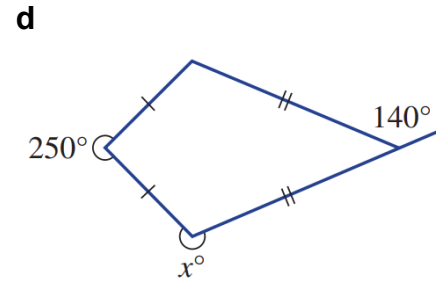
10 Determine the value of these pronumerals using your angle relationships knowledge.



$h = 70, f = 22.5, g = 110$

30

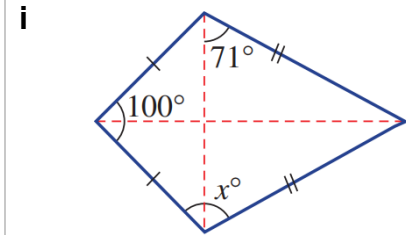
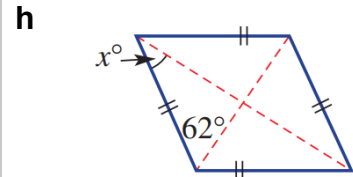
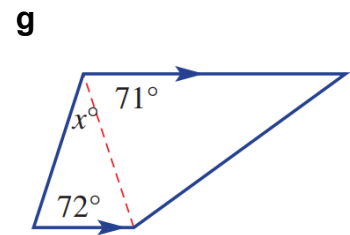
$y = 15, z = 40$



255

80

115

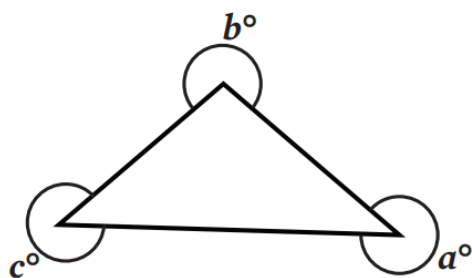


37

28

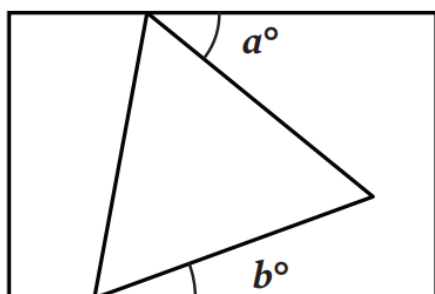
111

- 11 What is the value of  $a + b + c$ ? Explain your reasoning.



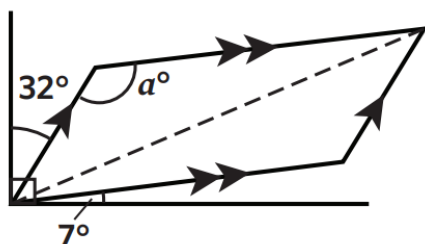
$$\begin{aligned} c + x &= 360 \\ b + y &= 360 \\ a + z &= 360 \\ c + x + b + y + a + z &= 1080 \\ a + b + c &= 1080 - (x + y + z) \\ x + y + z &= 180 \\ \therefore a + b + c &= 1080 - 180 = 900^\circ \end{aligned}$$

- 12 The diagram has an equilateral triangle inside a rectangle. Determine the value of  $a + b$ . Explain your reasoning.



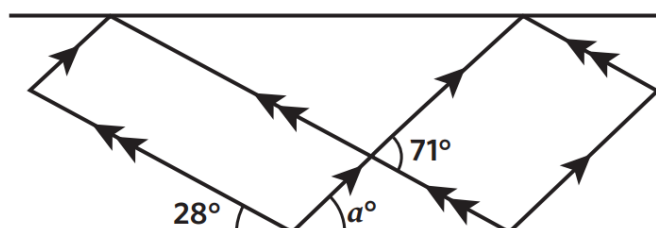
Construct line AB at C parallel to sides of rectangle  
 $x = a$  and  $y = b$  (alternate angles on parallel lines)  
 $x + y = 60^\circ$  (angles in equilateral  $\triangle$ )  
 $\therefore a + b = 60^\circ$

- 13 Determine the value of  $a$  in this diagram. Explain your reasoning.



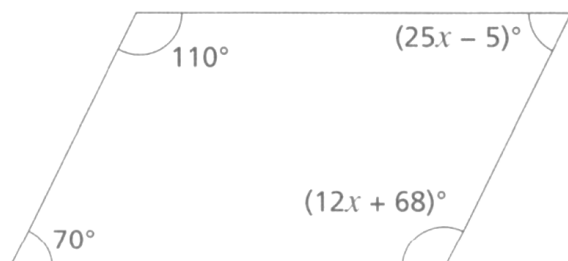
$$\begin{aligned} b &= 90 - 32 - 7 \text{ (angles in right-angle)} \\ &= 51^\circ \\ 2a + 2b &= 360 \text{ (angle sum of quad)} \\ 2a &= 360 - 2b = 360 - 2(51) \\ 2a &= 258^\circ \\ \therefore a &= 129^\circ \end{aligned}$$

- 14 Determine the value of  $a$  in this diagram. Explain your reasoning.



$$\begin{aligned} 2x + 71 + 71 &= 360 \text{ (angle sum of quad)} \\ 2x &= 218 \\ x &= 109 \\ a + 28 + 109 &= 180 \text{ (angles on straight line)} \\ a &= 43^\circ \end{aligned}$$

- 15 Is this shape a parallelogram? Explain your reasoning.

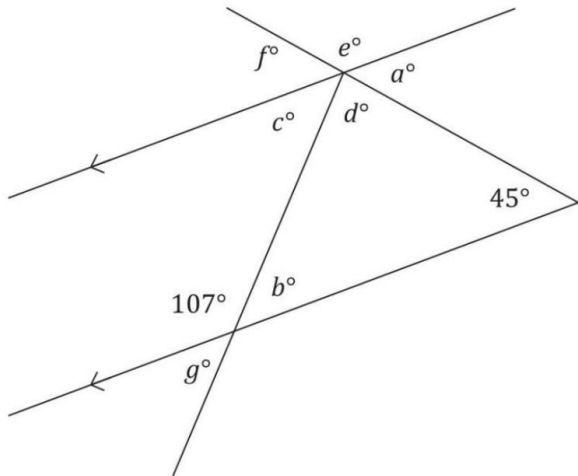


If parallelogram, then co-interior angles equal  
 $110 + 25x - 5 = 180$   
 $25x = 75$   
 $x = 3$   
 check using the other pair of co-interior angles  $70 + 12x + 68 = 180$   
 $70 + 12(3) + 68 = 174 \neq 180$   
 $\therefore$  not parallelogram

# Mixed Practice

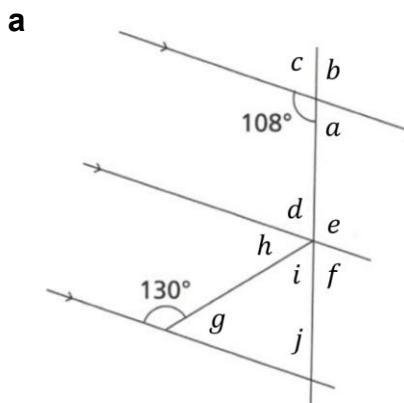
## FOUNDATION

1 Determine the value represented by each pronumeral.

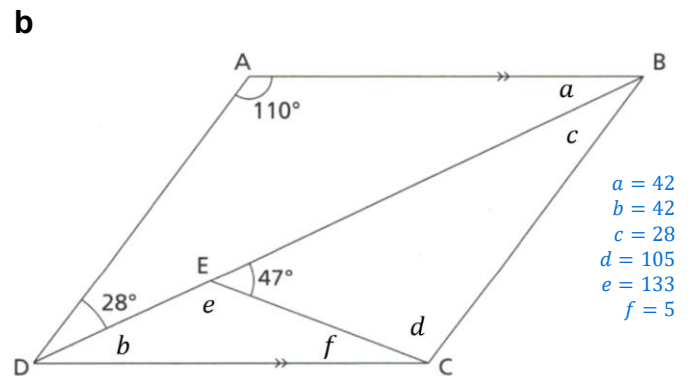


$$a = 45, b = 73, c = 73, d = 62, e = 135, f = 45, g = 73$$

2 Determine the sizes of all the unknown angles.

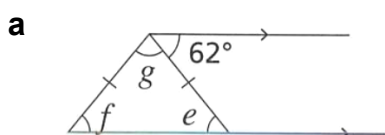


$$\begin{aligned} a &= 72 \\ b &= 108 \\ c &= 72 \\ d &= 72 \\ e &= 108 \\ f &= 72 \\ g &= 50 \\ h &= 50 \\ i &= 58 \\ j &= 72 \end{aligned}$$

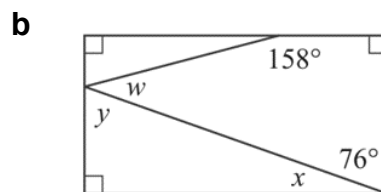


$$\begin{aligned} a &= 42 \\ b &= 42 \\ c &= 28 \\ d &= 105 \\ e &= 133 \\ f &= 5 \end{aligned}$$

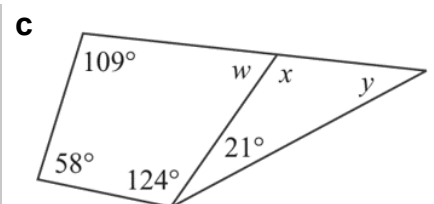
3 Work out the values of the pronumerals.



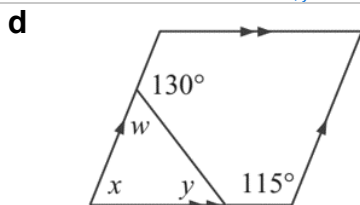
$$e = 62, f = 62, g = 56$$



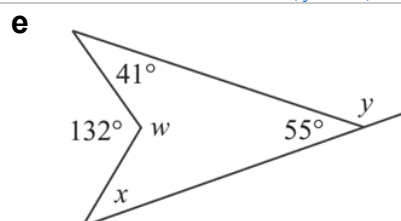
$$x = 14, y = 76, w = 36$$



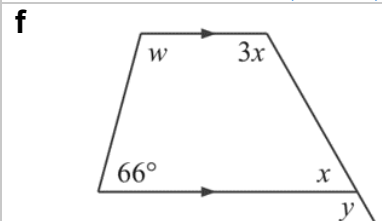
$$w = 69, x = 111, y = 48$$



$$w = 50, x = 65, y = 65$$



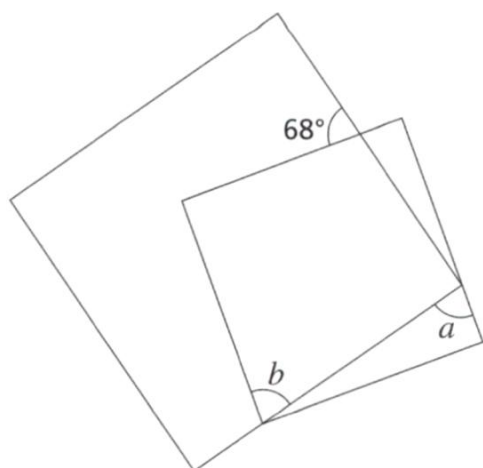
$$w = 228, x = 36, y = 125$$



$$x = 45, w = 114, y = 135$$

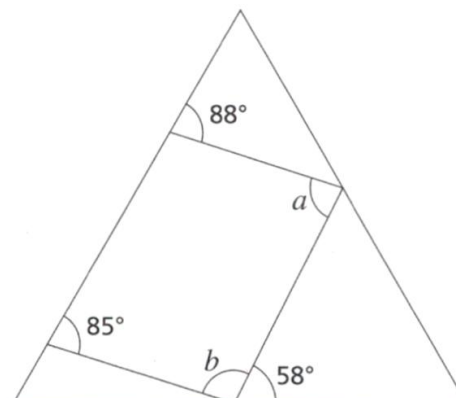
4 Work out the size of angles  $a$  and  $b$ .

a Two overlapping squares



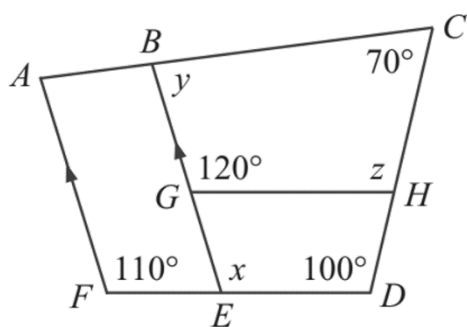
$$a = b = 68^\circ$$

b Quadrilateral inside an equilateral triangle



$$a = 86^\circ, b = 97^\circ$$

5 Find the values of the pronumerals. Provide full reasoning.



$$x = 110^\circ (\text{corresponding angles on parallel lines})$$

$$\angle EGH = 60^\circ (\text{angle sum of straight line})$$

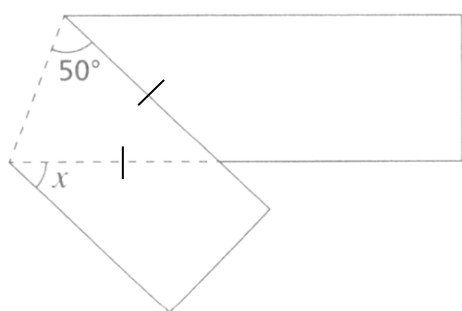
$$\angle GHD = 90^\circ (\text{angle sum of quadrilateral})$$

$$z = 90^\circ (\text{angle sum of straight line})$$

$$y = 80^\circ (\text{angle sum of quadrilateral})$$

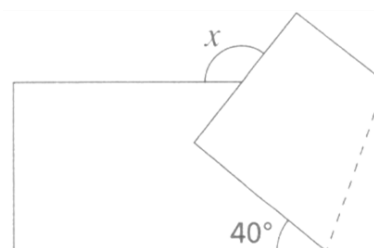
6 Juna is exploring angles by folding rectangular strips of paper. Work out the size of each angle marked  $x$ .

a



$$x = 80$$

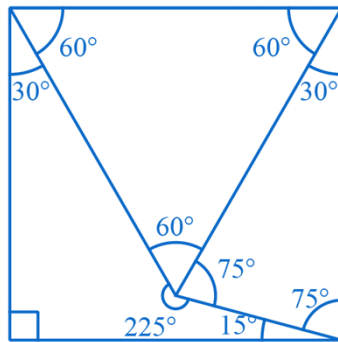
b



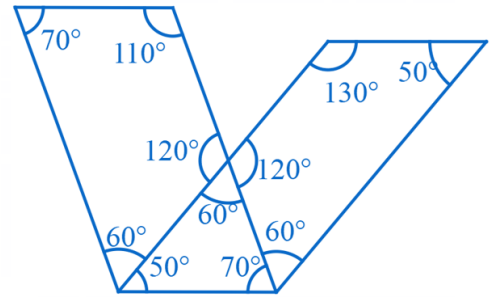
$$x = 130$$

7 Find every angle you can.

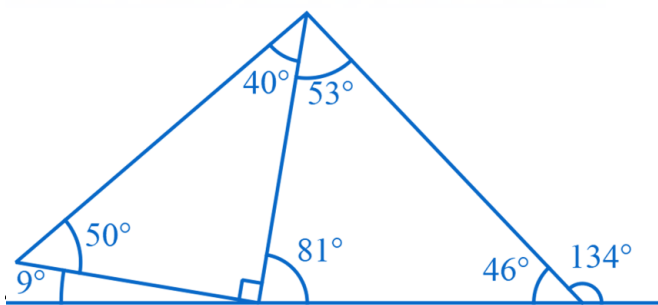
a PQRS is a square and PQT is an equilateral triangle.



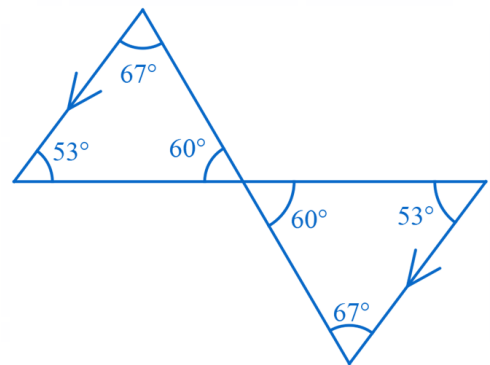
b This diagram shows two overlapping parallelograms.



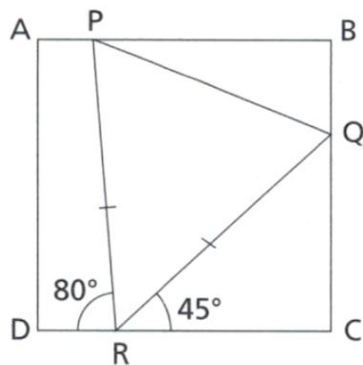
c



d



8 The diagram shows a triangle in a square. Find the size of  $\angle BQP$ . Provide full reasoning.



$$\angle PRQ = 55^\circ \text{ (angles on a straight line add up to } 180^\circ)$$

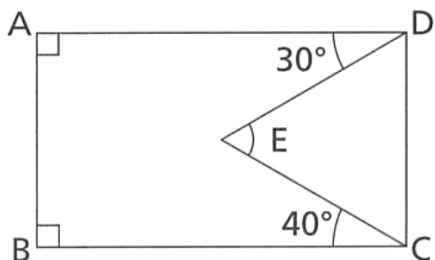
$$\angle RQP = 62.5^\circ \text{ (angles in a triangle add up to } 180^\circ \text{ and two angles in an isosceles triangle are equal)}$$

$$\angle CQR = 45^\circ \text{ (angles in a triangle add up to } 180^\circ)$$

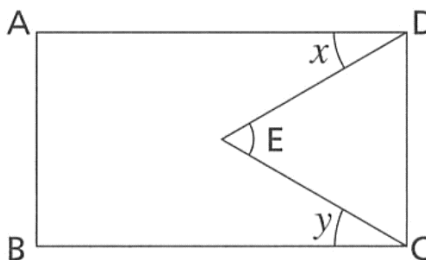
$$\angle BQP = 72.5^\circ \text{ (angles on a straight line add up to } 180^\circ)$$

9 In both diagrams, ABCD is a rectangle.

i



ii



a Using diagram i, prove that  $\angle DEC = 70^\circ$ . Provide full reasoning.

$$\angle EDC = 90^\circ - 30^\circ = 60^\circ \text{ (angles in a rectangle are right angles)}$$

$$\angle ECD = 90^\circ - 40^\circ = 50^\circ \text{ (angles in a rectangle are right angles)}$$

$$\angle DEC = 180^\circ - (60^\circ + 50^\circ) = 70^\circ \text{ (angles in a triangle add up to } 180^\circ)$$

b Using diagram ii, prove that  $\angle DEC = (x + y)$

$$\angle EDC = 90^\circ - x \text{ (angles in a rectangle are right angles)}$$

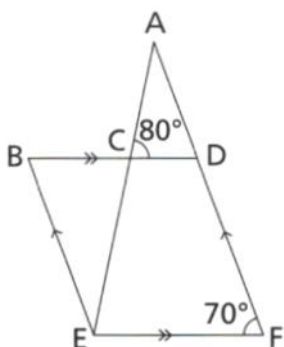
$$\angle ECD = 90^\circ - y \text{ (angles in a rectangle are right angles)}$$

$$\text{So } \angle EDC + \angle ECD = 180^\circ - (x + y)$$

$$\angle DEC = x + y \text{ (angles in a triangle add up to } 180^\circ)$$

10

a Prove that  $\angle BEC = 30^\circ$

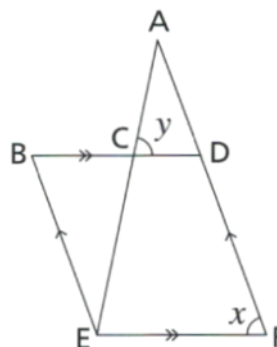


$$\angle BEF = 110^\circ \text{ (co-interior angles add up to } 180^\circ)$$

$$\angle AEF = 80^\circ \text{ (corresponding angles are equal)}$$

$$\angle BEC = 110^\circ - 80^\circ = 30^\circ$$

b Prove that  $\angle BEC = 180^\circ - (x + y)$

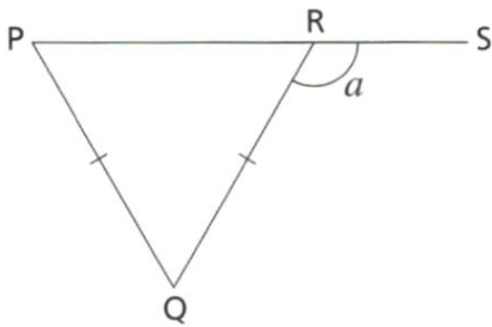


$$\angle BEF = 180^\circ - x \text{ (co-interior angles add up to } 180^\circ)$$

$$\angle AEF = y \text{ (corresponding angles are equal)}$$

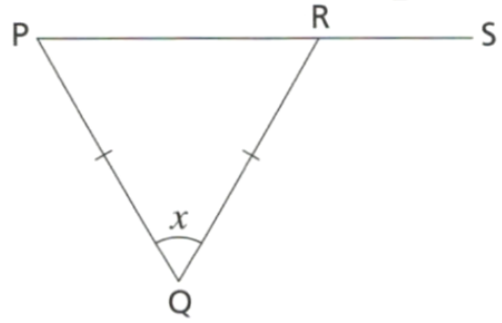
$$\angle BEC = (180^\circ - x) - y = 180^\circ - (x + y)$$

**a** Prove that that  $\angle PQR = (2a - 180)$



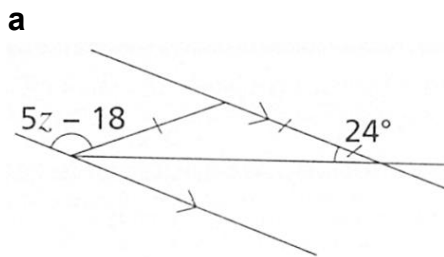
$$\begin{aligned}\angle PRQ &= 180^\circ - a \text{ (angles on a straight line add up to } 180\text{)} \\ \angle QPR &= 180^\circ - a \text{ (base angles in an isosceles triangle are equal)} \\ \angle PQR &= 180^\circ - (180^\circ - a + 180^\circ - a) \\ &= (2a - 180)^\circ \text{ (angles in a triangle add up to } 180^\circ\text{)}\end{aligned}$$

**b** Show that  $\angle QRS = \left(90 + \frac{x}{2}\right)$

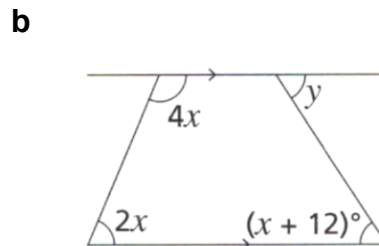


$$\begin{aligned}\angle PRQ &= \frac{180-x}{2} \text{ (base angles in an isosceles triangle are equal)} \\ \angle QRS &= 180 - \frac{180-x}{2} \text{ (angles on a straight line add up to } 180^\circ\text{)} \\ &= 90 + \frac{x}{2}\end{aligned}$$

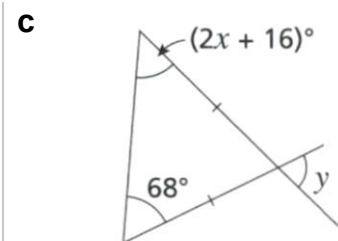
**12** In each diagram, work out the value of each pronumeral.



$$z = 30$$

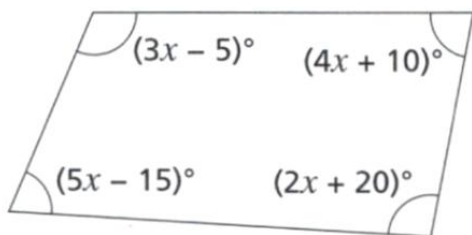


$$x = 30^\circ, y = 42^\circ$$



$$x = 26^\circ, y = 44^\circ$$

- 13 Show that this shape is a parallelogram. Justify your answer clearly.



Using angle sum of quadrilateral,  $x = 25$

If this is a parallelogram, then  $5x - 15 + 2x + 20 = 180$

$$\text{And } 3x - 5 + 4x + 10 = 180$$

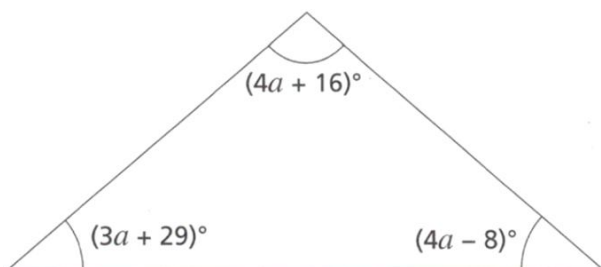
$$\text{And } 3x - 5 + 5x - 15 = 180$$

$$\text{And } 4x + 10 + 2x + 20 = 180$$

Because they would be co-interior.

By substituting  $x = 25$ , all these equations are true, so there are two pairs of parallel sides, and therefore the quadrilateral is a parallelogram.

- 14 Show that this triangle is isosceles.



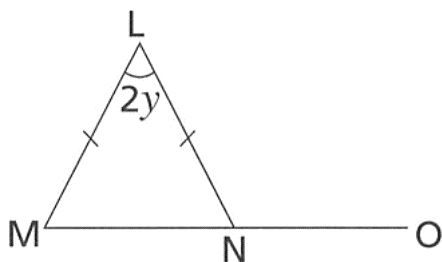
Using angle sum of triangle,  $a = 13$

$$3a + 29 = 3(13) + 29 = 68$$

$$4a + 16 = 4(13) + 16 = 68$$

Therefore isosceles as it has two equal angles.

- 15 Prove that  $\angle LNO = 90^\circ + y$



$$\angle LNM = \frac{180 - 2y}{2} \text{ (angles at the base of an isosceles triangle are equal)}$$

$$\text{So } \angle LNM = 90^\circ - y$$

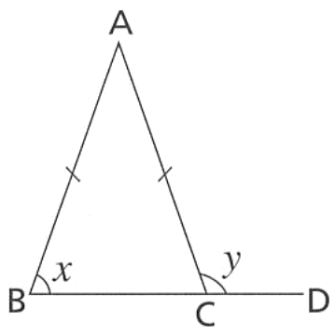
$$\angle LNM + \angle LNO = 180^\circ$$

(adjacent angles on a straight line add up to  $180^\circ$ )

$$\text{So } 90^\circ - y + \angle LNO = 180^\circ,$$

$$\text{so } \angle LNO = 90^\circ + y$$

16 Consider the diagram to the right.



a Prove that  $y = 180^\circ - x$

$\angle ACB = x$  (angles at the base of an isosceles triangle are equal)

$$\angle ACB + \angle ACD = 180^\circ$$

(adjacent angles on a straight line add up to  $180^\circ$ )

$$\text{So } x + y = 180^\circ, \text{ so } y = 180^\circ - x$$

b Prove that  $\angle BAC = 180^\circ - 2x$

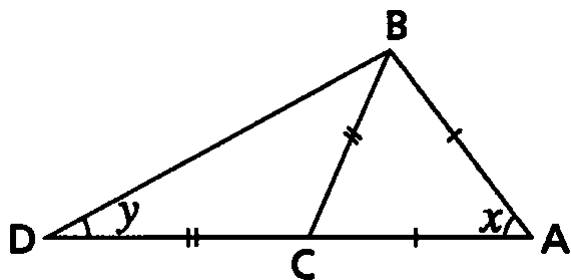
$$\angle BAC + \angle ABC + \angle ACB = 180^\circ$$

(angles in a triangle add up to  $180^\circ$ )

$$\text{So } \angle BAC + x + x = 180^\circ,$$

$$\text{so } \angle BAC = 180^\circ - 2x$$

17 Prove that  $x + 4y = 180^\circ$



$\angle CBD = y$  (angles at the base of an isosceles triangle are equal)

$$\angle BCD = 180^\circ - 2y \text{ (angles in a triangle add up to } 180^\circ)$$

$$\angle BCA = 2y \text{ (adjacent angles on a straight line add up to } 180^\circ)$$

$\angle ABC = 2y$  (angles at the base of an isosceles triangle are equal)

In triangle ABC,  $\angle BCA + \angle ABC +$

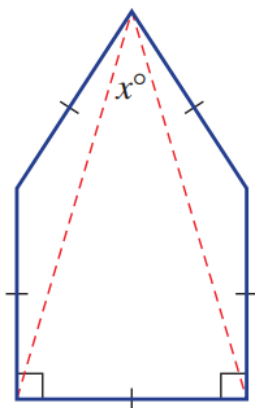
$$\angle CAB = 180^\circ \text{ (angles in a triangle add up to } 180^\circ)$$

$$\text{So } 2y + 2y + x = 180^\circ,$$

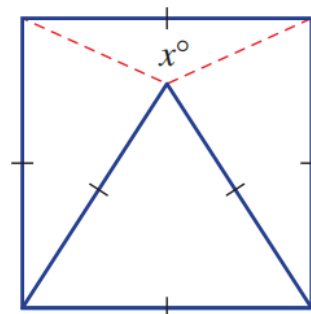
$$\text{hence } x + 4y = 180^\circ$$

18 Find the value of  $x$  in these diagrams.

a



b



30°

150°